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Chapter 1

Introduction

PLACEHOLDER

Chapter 2

Viscoelastic Fluid Mechanics and the Four-Roll Mill

The hydrodynamics of dilute and semi-dilute polymer solutions fundamentally departs from classical Newtonian fluid behavior. Macromolecular coils dissolved in a solvent undergo continuous conformational changes under applied flow fields, stretching along extensional trajectories and storing elastic energy with a characteristic relaxation memory. Consequently, the local stress state in polymeric fluids depends not merely on the instantaneous rate of deformation, but integrates the entire kinematic history experienced by fluid elements. Accurately modeling such transport phenomena requires a continuum mechanics formulation where universal conservation laws are coupled with non-linear, frame-indifferent constitutive equations.

In this chapter, we develop the continuum framework governing viscoelastic flows within the four-roll mill geometry. Starting from the foundational balance laws of mass and linear momentum, we introduce the kinematic reductions characteristic of confined, low-inertia polymeric flows. We then establish the constitutive formulation of viscoelasticity through the Upper-Convected Time Derivative, analyzing both the classical Oldroyd-B model, including the physical roots of its

extensional stress singularity, and non-linear stress-saturating closures such as the Giesekus and Phan-Thien-Tanner (PTT) formulations. Finally, we cast the governing system into dimensionless form using a viscous stress scale, detail the geometry and hyperbolic stagnation kinematics of the four-roll mill, and prescribe the boundary constraints that establish the boundary value problem.

2.1 General Conservation Laws

From a macroscopic continuum perspective, fluid motion is governed by the universal conservation laws of mass and linear momentum [1].

2.1.1 Mass Balance

For a fluid possessing local density $\rho(\boldsymbol{x}, t)$, the conservation of mass over an arbitrary control volume is expressed by the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{u}) = 0. \tag{2.1}$$

2.1.2 Momentum Balance

Applying Newton's second law of motion to an infinitesimal fluid element yields the local momentum balance:

$$\rho \frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} = \nabla \cdot \boldsymbol{\sigma} + \rho \boldsymbol{g}, \tag{2.2}$$

where $\boldsymbol{g}$ represents external body force accelerations and $\boldsymbol{\sigma}$ denotes the symmetric total stress tensor. The left-hand side operator $\frac{\mathrm{D}}{\mathrm{D}t}$ is the substantial (or material) derivative, defined as:

$$\frac{\mathrm{D}\boldsymbol{u}}{\mathrm{D}t} \equiv \frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u}. \tag{2.3}$$

In the Eulerian framework, the substantial derivative expresses the total rate of velocity change experienced by a moving fluid parcel, including the time derivative $\frac{\partial \boldsymbol{u}}{\partial t}$ at a fixed spatial position and the convective part $(\boldsymbol{u} \cdot \nabla)\boldsymbol{u}$. Expanding Equation (2.2):

$$\rho \left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} \right) = \nabla \cdot \boldsymbol{\sigma} + \rho \boldsymbol{g}. \tag{2.4}$$

In order to separate isotropic hydrostatic effects from deviatoric shearing and stretching deformations, the total stress tensor $\boldsymbol{\sigma}$ is split into an isotropic pressure contribution and an extra-stress tensor $\boldsymbol{T}$:

$$\boldsymbol{\sigma} = -p\boldsymbol{I} + \boldsymbol{T}, \tag{2.5}$$

where p is the isotropic hydrodynamic pressure, $\boldsymbol{I}$ is the identity tensor, and $\boldsymbol{T}$ represents all viscous and viscoelastic stresses. Substituting Equation (2.5) into Equation (2.4):

$$\rho \left(\frac{\partial \boldsymbol{u}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{u} \right) = -\nabla p + \nabla \cdot \boldsymbol{T} + \rho \boldsymbol{g}. \tag{2.6}$$

2.2 Physical Assumptions

The general conservation equations can be simplified by introducing the scale-appropriate physical hypotheses of the four-roll mill:

1. **Incompressibility**: Under moderate operating pressures and low characteristic velocities, density variations are negligible ($\rho = \text{const}$). The continuity equation (2.1) simplifies to the divergence-free condition:

$$\nabla \cdot \boldsymbol{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0. \tag{2.7}$$

2. **Steady-state regime**: We focus on the fully established operating regime,

setting all explicit time derivatives to zero:

$$\frac{\partial(\cdot)}{\partial t} = \mathbf{0}. \tag{2.8}$$

3. **Absence of body forces**: The four-roll mill operates in a horizontal cross-sectional plane (x, y). Because gravity acts along the vertical cylinder axis (z), perpendicular to the flow plane, external body forces vanish: $\boldsymbol{g} = \mathbf{0}$.

4. **Isothermal operating conditions**: Because velocities are low and viscous heating is negligible, the temperature field remains uniform. Fluid transport properties, including viscosities and relaxation times, are therefore treated as constants.

5. **Two-dimensional planar kinematics**: Given the large aspect ratio of the cylindrical rollers along the z-axis, three-dimensional end effects are negligible in the transverse mid-plane. The velocity field is modeled as strictly planar:

$$\boldsymbol{u} = \begin{pmatrix} u(x,y) \\ v(x,y) \\ 0 \end{pmatrix}. \tag{2.9}$$

2.2.1 Stream Function Formulation

In two-dimensional incompressible flow, the solenoidal constraint $\nabla \cdot \boldsymbol{u} = 0$ can be satisfied directly by defining a scalar stream function $\psi(x, y)$ such that:

$$u = \frac{\partial \psi}{\partial y}, \qquad v = -\frac{\partial \psi}{\partial x}. \tag{2.10}$$

Because mixed partial derivatives commute ($\frac{\partial^2\psi}{\partial x\partial y} = \frac{\partial^2\psi}{\partial y\partial x}$), the velocity divergence vanishes identically:

$$\nabla \cdot \boldsymbol{u} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = \frac{\partial^2\psi}{\partial x\partial y} - \frac{\partial^2\psi}{\partial y\partial x} \equiv 0. \tag{2.11}$$

Parameterizing the velocity through ψ satisfies mass conservation everywhere by construction, replacing the two velocity components (u, v) with a single scalar field.

Under these physical assumptions and kinematically reduced velocity representation, the steady balance of linear momentum simplifies to:

$$\rho(\boldsymbol{u} \cdot \nabla\boldsymbol{u}) = -\nabla p + \nabla \cdot \boldsymbol{T}. \tag{2.12}$$

2.3 Constitutive Modeling

In two dimensions, the momentum balance (2.12) contains six unknown scalar fields $(u, v, p, T_{xx}, T_{xy}, T_{yy})$, while mass and momentum conservation provide only three equations. A constitutive equation for the extra-stress tensor $\boldsymbol{T}$ is required to close the system.

For dilute and semi-dilute polymer solutions, we split the extra-stress tensor $\boldsymbol{T}$ into a Newtonian solvent contribution and an elastic polymeric contribution $\boldsymbol{\tau}$ [1, 2]:

$$\boldsymbol{T} = 2\mu_s\boldsymbol{D} + \boldsymbol{\tau}, \tag{2.13}$$

where μ_s is the dynamic viscosity of the Newtonian solvent, and $\boldsymbol{D}$ is the symmetric rate-of-strain tensor:

$$\boldsymbol{D} = \frac{1}{2}\left[\nabla\boldsymbol{u} + (\nabla\boldsymbol{u})^T\right]. \tag{2.14}$$

Taking the divergence of the solvent stress and leveraging incompressibility ($\nabla \cdot \boldsymbol{u} = 0$):

$$\nabla \cdot (2\mu_s \boldsymbol{D}) = \mu_s \nabla \cdot \left(\nabla \boldsymbol{u} + (\nabla \boldsymbol{u})^T\right) = \mu_s \left(\nabla^2 \boldsymbol{u} + \nabla(\nabla \cdot \boldsymbol{u})\right) = \mu_s \nabla^2 \boldsymbol{u}. \quad (2.15)$$

Substituting Equation (2.15) into Equation (2.12) yields the steady momentum equation:

$$\rho(\boldsymbol{u} \cdot \nabla \boldsymbol{u}) = -\nabla p + \mu_s \nabla^2 \boldsymbol{u} + \nabla \cdot \boldsymbol{\tau}. \quad (2.16)$$

2.3.1 Upper-Convected Time Derivative

Polymer chains suspended in a solvent stretch, orient, and relax over finite timescales. Because fluid parcels translate, rotate, and deform as they move through the domain, constitutive relations require a rate of stress that is invariant under changes of reference frame (material frame-indifference). A standard time derivative evaluated at a fixed point would interpret rigid-body rotation or parcel stretching as a change in stress, even if the internal state of the polymer remained unaltered.

To measure true rheological evolution, the time rate of change must be evaluated in a local coordinate system that deforms affinely with the fluid element. This objective tensor rate is the *Upper-Convected Time Derivative* (UCTD), denoted by $\overset{\nabla}{\boldsymbol{\tau}}$:

$$\overset{\nabla}{\boldsymbol{\tau}} \equiv \frac{\partial \boldsymbol{\tau}}{\partial t} + \boldsymbol{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \boldsymbol{u})\boldsymbol{\tau} - \boldsymbol{\tau}(\nabla \boldsymbol{u})^T, \quad (2.17)$$

where $(\nabla \boldsymbol{u})_{ij} = \partial u_i / \partial x_j$. For steady flows ($\partial \boldsymbol{\tau}/\partial t = \boldsymbol{0}$), the operator combines advective stress transport along streamlines, $\boldsymbol{u} \cdot \nabla \boldsymbol{\tau}$, with kinematic stretching and rotation induced by local velocity gradients, $-(\nabla \boldsymbol{u})\boldsymbol{\tau} - \boldsymbol{\tau}(\nabla \boldsymbol{u})^T$.

2.3.2 Oldroyd-B Constitutive Model

The Oldroyd-B model [4] provides the standard starting point for viscoelastic fluid modeling. It idealizes macromolecules as non-interacting Hookean dumbbells suspended in a Newtonian solvent [1]. Under steady conditions, the polymeric extra-stress tensor $\boldsymbol{\tau}$ satisfies:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} = 2\mu_p \boldsymbol{D}, \tag{2.18}$$

where λ is the relaxation time and μ_p is the polymeric viscosity. The total zero-shear dynamic viscosity of the solution is:

$$\mu_0 = \mu_s + \mu_p. \tag{2.19}$$

In steady simple shear flow, Oldroyd-B predicts a constant viscosity $\eta(\dot{\gamma}) = \mu_0$, reproducing the behavior of Boger fluids.

Planar Extensional Behavior

In strong extensional flows, the Oldroyd-B model exhibits an unphysical stress singularity. For a steady planar elongational flow centered at the origin, $u = Ex$ and $v = -Ey$ with extension rate $E > 0$, the tensile normal stress along the stretching axis is:

$$\tau_{xx} = \frac{2\mu_p E}{1 - 2\lambda E}. \tag{2.20}$$

As the extension rate approaches the critical threshold:

$$E_{\text{crit}} = \frac{1}{2\lambda}, \tag{2.21}$$

the denominator vanishes. Because Hookean springs have infinite extensibility, the polymer chains can stretch without bound, driving both τ_{xx} and the exten-

sional viscosity η_E to infinity.

This stress singularity near $2\lambda E \to 1$ is the constitutive origin of the *High-Weissenberg Number Problem* (HWNP). Near extensional stagnation points, it generates steep stress layers that cause numerical instability in standard mesh-based solvers [3].

2.3.3 Giesekus Model

To overcome the unphysical stress singularity of Oldroyd-B, Giesekus [1, 20] introduced a quadratic non-linear stress contribution derived from configuration-dependent hydrodynamic drag. In semi-dilute solutions, stretching macromolecular chains align parallel to each other, reducing drag along the orientation direction relative to the transverse directions. This anisotropic friction is incorporated by modifying the stress relaxation rate:

$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha\lambda}{\mu_p}\boldsymbol{\tau}^2 = 2\mu_p \boldsymbol{D}, \tag{2.22}$$

where $\alpha \in [0, 0.5]$ is the dimensionless mobility parameter, and $\boldsymbol{\tau}^2 = \boldsymbol{\tau} \cdot \boldsymbol{\tau}$. When $\alpha = 0$, isotropic drag is recovered and the model reduces identically to Oldroyd-B. For $\alpha > 0$, the quadratic dissipation term $\boldsymbol{\tau}^2$ provides two key rheological enhancements:

1. **Bounded extensional plateau**: Extensional stresses no longer diverge at E_{crit}; instead, η_E reaches an asymptotic finite plateau at high strain rates, completely resolving the constitutive singularity.

2. **Shear-thinning behavior**: The steady shear viscosity $\eta(\dot{\gamma})$ decreases monotonically with increasing shear rate, closely matching experimental polymer melt and concentrated solution data.

2.3.4 Linear Phan-Thien-Tanner (PTT) Model

Derived from the transient network theory of concentrated polymer solutions and melts [1,21], the Phan-Thien-Tanner (PTT) model treats macromolecules as a network of temporary entanglements. Under strong deformation, mechanical tension on the network strands accelerates junction destruction. In its linear formulation, this stress-induced network breakdown is modeled by scaling the linear relaxation term:

$$f(\boldsymbol{\tau})\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} = 2\mu_p \boldsymbol{D}, \tag{2.23}$$

where the scalar function $f(\boldsymbol{\tau})$ is defined as:

$$f(\boldsymbol{\tau}) = 1 + \frac{\varepsilon\lambda}{\mu_p}\mathrm{tr}(\boldsymbol{\tau}), \tag{2.24}$$

with $\mathrm{tr}(\boldsymbol{\tau}) = \tau_{xx} + \tau_{yy}$ (since $\tau_{zz} = 0$ under planar deformation for linear PTT) and $\varepsilon \geq 0$ representing the dimensionless extensibility parameter. When $\varepsilon = 0$, $f(\boldsymbol{\tau}) \equiv 1$ and the linear Oldroyd-B constitutive law is recovered. For $\varepsilon > 0$, high trace stresses accelerate stress relaxation, bounding extensional viscosity growth and preventing unbounded coil stretch near extensional stagnation points.

2.4 Dimensionless Governing Equations

Nondimensionalization scales physical variables to order unity and establishes the governing dimensionless groups [3]. In low-Reynolds viscoelastic flows, scaling by a viscous stress unit rather than an inertial dynamic pressure improves numerical conditioning and balances the magnitudes of the governing differential operators.

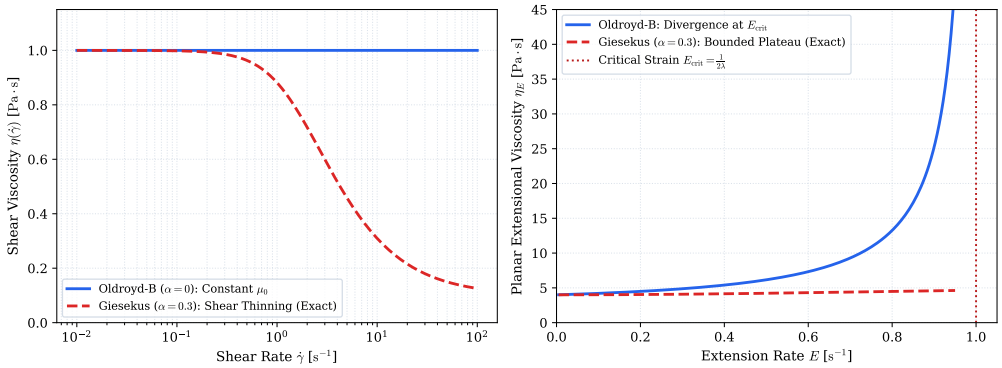


Figure 2.1: Example comparison of rheological model responses: (a) steady shear viscosity $\eta(\dot{\gamma})$ as a function of shear rate, illustrating the constant Newtonian-like plateau of Oldroyd-B ($\alpha = 0$) versus the non-linear shear-thinning response of Giesekus ($\alpha > 0$); (b) planar extensional viscosity $\eta_E(E)$ as a function of extension rate, contrasting the unphysical asymptotic divergence of Oldroyd-B at the critical strain threshold $E_{\text{crit}} = 1/(2\lambda)$ with the bounded saturation plateau provided by non-linear closures.

2.4.1 Reference Scales and Dimensionless Parameters

We choose reference length H_{ref} (the roller radius R), reference velocity U_{ref} (the roller surface peripheral speed u_{roll}), and a characteristic dynamic viscosity scale η_0 (taken as the baseline total viscosity μ_0). The dimensionless variables are defined as:

$$\boldsymbol{x}^* = \frac{\boldsymbol{x}}{H_{\text{ref}}}, \quad \boldsymbol{u}^* = \frac{\boldsymbol{u}}{U_{\text{ref}}}, \quad p^* = \frac{p}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}, \quad \boldsymbol{\tau}^* = \frac{\boldsymbol{\tau}}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}. \tag{2.25}$$

Dropping asterisks for notational simplicity, the flow regime is governed by three fundamental dimensionless groups:

$$Re = \frac{\rho U_{\text{ref}} H_{\text{ref}}}{\eta_0} \qquad \text{(Reynolds number)}, \tag{2.26}$$

$$Wi = \frac{\lambda U_{\text{ref}}}{H_{\text{ref}}} \qquad \text{(Weissenberg number)}, \tag{2.27}$$

$$\beta = \frac{\mu_s}{\mu_0} = \frac{\mu_s}{\mu_s + \mu_p} \qquad \text{(Viscosity ratio)}. \tag{2.28}$$

Each dimensionless group carries a precise physical meaning:

- **Reynolds number (Re)**: Compares convective inertial forces to viscous dissipation forces. In the present benchmark, $Re \ll 1$, placing the system firmly in the laminar creeping flow (Stokes) regime.

- **Weissenberg number (Wi)**: Compares the characteristic elastic relaxation time of the polymer λ to the characteristic kinematic timescale of the flow $H_{\text{ref}}/U_{\text{ref}}$. It quantifies the degree of flow-induced macromolecular orientation and elastic non-linearity.

- **Viscosity ratio (β)**: Measures the fraction of total viscosity contributed by the Newtonian solvent. The limits $\beta = 1$ and $\beta \to 0$ correspond to a purely Newtonian fluid and a solvent-free polymer melt, respectively. The dimensionless solvent and polymeric viscosities are given by $\mu_s^* = \beta$ and $\mu_p^* = 1 - \beta$.

2.4.2 Dimensionless Governing System

Expressed in terms of dimensionless variables, the steady planar governing equations become:

$$\nabla \cdot \boldsymbol{u} = 0, \tag{2.29}$$

$$Re(\boldsymbol{u} \cdot \nabla \boldsymbol{u}) + \nabla p - \beta \nabla^2 \boldsymbol{u} - \nabla \cdot \boldsymbol{\tau} = \boldsymbol{0}, \tag{2.30}$$

$$\boldsymbol{\mathcal{G}}(\boldsymbol{\tau}) + Wi\left(\boldsymbol{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \boldsymbol{u})\boldsymbol{\tau} - \boldsymbol{\tau}(\nabla \boldsymbol{u})^T\right) - 2(1-\beta)\boldsymbol{D} = \boldsymbol{0}, \tag{2.31}$$

where the constitutive operator $\boldsymbol{\mathcal{G}}(\boldsymbol{\tau})$ provides a unified closure for the models under investigation:

$$\boldsymbol{\mathcal{G}}(\boldsymbol{\tau}) = \begin{cases} \boldsymbol{\tau} & \text{(Oldroyd-B)}, \\ \boldsymbol{\tau} + \frac{\alpha Wi}{1-\beta}\boldsymbol{\tau}^2 & \text{(Giesekus)}, \\ \left(1 + \frac{\varepsilon Wi}{1-\beta}\mathrm{tr}(\boldsymbol{\tau})\right)\boldsymbol{\tau} & \text{(Linear PTT)}. \end{cases} \tag{2.32}$$

2.5 Four-Roll Mill

The four-roll mill is a classic benchmark designed to generate controlled two-dimensional planar extensional flow with a central stagnation point [5–7, 17]. Originally introduced by Taylor [5] and systematically analyzed by Fuller and Leal [7, 17], it provides an ideal apparatus for investigating macromolecular deformation away from solid walls. As illustrated conceptually in Figure 2.2, the setup comprises four cylindrical rollers placed symmetrically inside a closed square cavity.

2.5.1 Flow Index Parameter

In the central region bounded by the four rollers, the velocity field can be represented as a linear superposition of extension and solid-body rotation [7, 17]:

$$\boldsymbol{u}(x, y) = \begin{pmatrix} Ex + \omega y \\ -Ey - \omega x \end{pmatrix}, \tag{2.33}$$

where $E \geq 0$ is the extension rate and $\omega \geq 0$ is the angular rotation rate. To quantify the relative magnitude of extension versus rotation, we define a dimen-

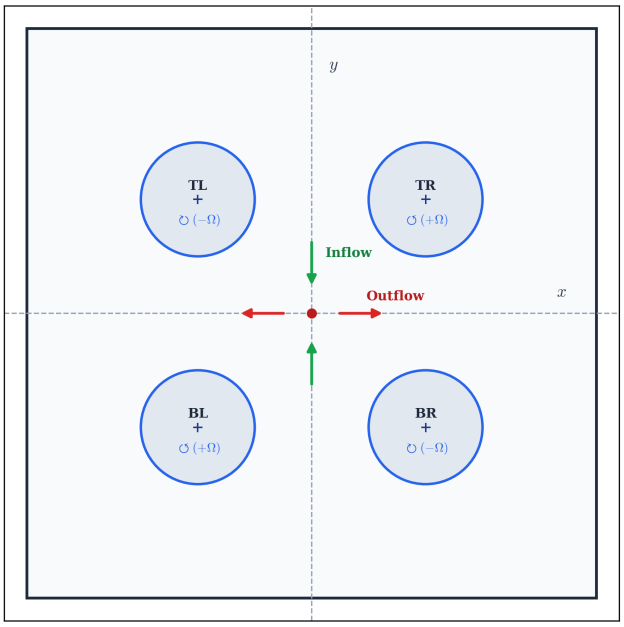


Figure 2.2: Conceptual diagram of the four-roll mill geometry, illustrating the four rotating cylinders (adjacent rolls counter-rotate with angular velocity Ω, diagonal rolls co-rotate), the central hyperbolic stagnation point $(0, 0)$, and the resulting elongational inflow and outflow directions.

sionless flow type parameter $\chi \in [-1, 1]$:

$$\chi = \frac{E - \omega}{E + \omega}. \tag{2.34}$$

Depending on roller rotation speeds and directions, this local kinematics covers three canonical flow regimes (in Figure 2.3):

- $\chi = +1$ $(\omega = 0)$: **Pure Planar Extensional Flow**. Generated when adjacent rollers counter-rotate symmetrically (diagonal rollers co-rotate). Fluid elements experience pure elongation along x and compression along y, producing hyperbolic streamlines $xy = \text{const}$.

- $\chi = 0$ $(E = \omega)$: **Simple Shear Flow**. Characterized by equal strain and rotation rates, producing rectilinear shear streamlines locally near the origin.

- $\chi = -1$ ($E = 0$): **Pure Solid-Body Rotation**. Generated when rollers rotate in unison to drive bulk cavity circulation, resulting in concentric circular streamlines without stretching.

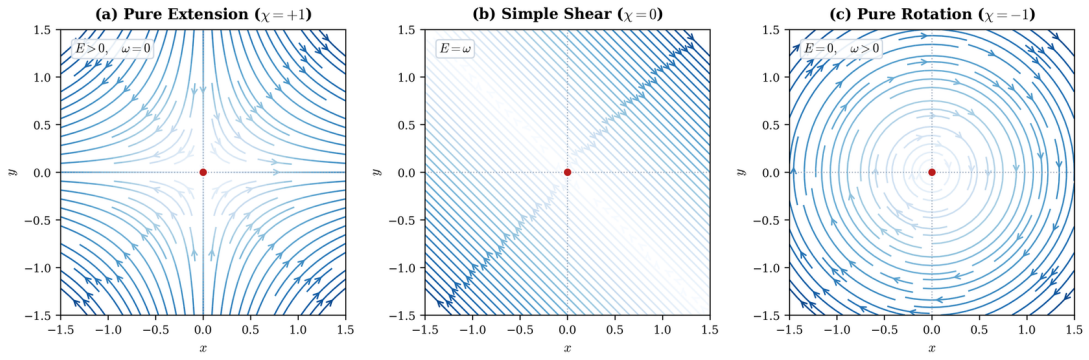


Figure 2.3: Streamline patterns of the two-dimensional linear flow field in Equation (2.33) for the three canonical regimes of the flow classification parameter χ: (a) pure planar extension ($\chi = +1$), (b) simple shear ($\chi = 0$), and (c) pure solid-body rotation ($\chi = -1$). The red marker indicates the central stagnation point $(0, 0)$.

2.5.2 Stagnation Point

In the planar extensional regime ($\chi = +1$), the velocity vanishes identically at the cavity center:

$$\boldsymbol{u}(0, 0) = \mathbf{0}. \tag{2.35}$$

Along the vertical inflow streamline ($x = 0$), incoming fluid parcels decelerate toward zero velocity, reaching asymptotically long residence times near the origin. This kinematics allows elastic strain to accumulate over durations comparable to or exceeding the polymer relaxation time λ, making the four-roll mill an ideal benchmark for probing strong extensional stresses far from solid walls.

2.5.3 Geometric Setup and Operating Parameters

Following the established benchmark setup of D'Avino et al. [18], we define the physical dimensions and operating conditions of the apparatus:

- **Square cavity**: Side length $L = 50\,\text{mm}$ spanning the domain $[-25, 25]\,\text{mm}$.
- **Cylindrical rollers**: Radius $R = 5\,\text{mm}$ ($H_{\text{ref}} = R = 0.1L$).
- **Roller centers**: Located symmetrically at $(\pm 10, \pm 10)\,\text{mm}$ ($d_{\text{roll}} = 10\,\text{mm}$).
- **Roller surface speed**: Peripheral velocity $u_{\text{roll}} = \Omega R = 8.333 \times 10^{-3}\,\text{m/s}$.
- **Dimensionless domain**: In units of $H_{\text{ref}} = R$, the computational domain spans $[-5, 5]^2$, with roller centers located at $(\pm 2.0, \pm 2.0)$ and non-dimensional roller radius $R^* = 1.0$.

As shown in Figure 2.4, these dimensions define the operational space and solid boundaries of the cavity.

The geometric, fluid physical, and characteristic dimensionless parameters investigated in this work are organized systematically in Table 2.1.

2.5.4 Boundary Conditions

To complete the mathematical formulation, boundary conditions are imposed along allrelevant domain boundaries:

1. **No-Slip on Roller Surfaces (Γ_{rolls})**: On the boundary of each cylinder $k \in \{\text{TL}, \text{TR}, \text{BL}, \text{BR}\}$, the fluid velocity satisfies the no-slip Dirichlet condition with specified angular rotation:

$$\boldsymbol{u}|_{\Gamma_k} = \boldsymbol{\Omega}_k \times (\boldsymbol{x} - \boldsymbol{x}_k), \tag{2.36}$$

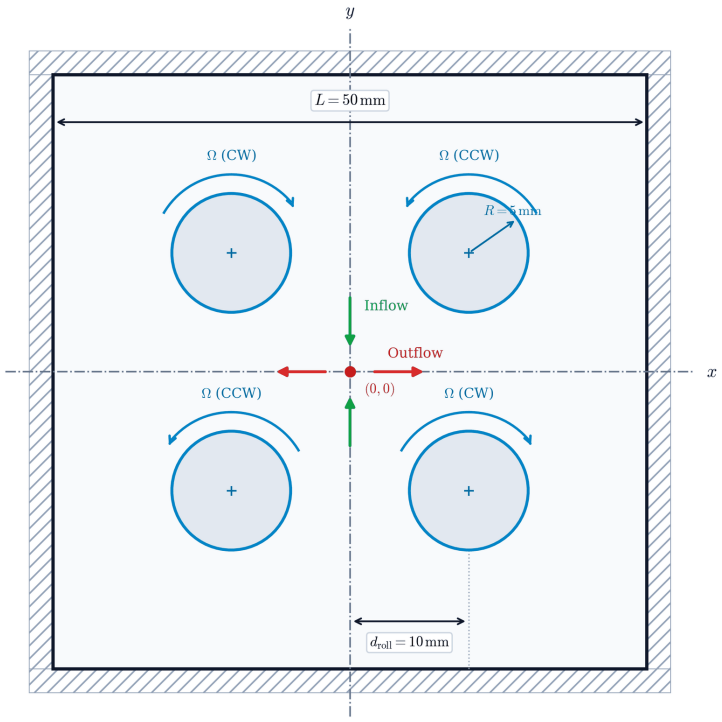


Figure 2.4: Detailed geometric dimensions and coordinate setup of the four-roll mill: square cavity side $L = 50\,\text{mm}$, cylinder radii $R = 5\,\text{mm}$, roller center offset $d_{\text{roll}} = 10\,\text{mm}$, Cartesian axes (x, y), and peripheral rotation directions with angular velocity Ω.

Table 2.1: Geometric, physical, and dimensionless parameters of the four-roll mill benchmark.

Parameter	**Symbol**	**Value / Range**	**Unit**
Geometric and Kinematic Setup			
Cavity side length	L	0.05	m
Roller radius (reference length)	$H_{\text{ref}} = R$	0.005	m
Roller center offset	d_{roll}	0.010	m
Roller peripheral velocity	$u_{\text{roll}} = U_{\text{ref}}$	8.333×10^{-3}	m/s
Fluid Physical Properties			
Fluid density	ρ	1000	kg/m^3
Characteristic viscosity scale	$\eta_0 = \mu_0$	1.0	$\text{Pa} \cdot \text{s}$
Solvent dynamic viscosity	μ_s	0.1–0.5	$\text{Pa} \cdot \text{s}$
Polymer dynamic viscosity	μ_p	0.5–0.9	$\text{Pa} \cdot \text{s}$
Polymer relaxation time	λ	0.05–1.0	s
Dimensionless Governing Groups			
Reynolds number	Re	$0.0417 \ll 1$	–
Weissenberg number	Wi	0.083–1.667	–
Viscosity ratio	β	0.1–0.5	–

where $\boldsymbol{x}_k$ is the roller center and the angular velocity vector is directed along the out-of-plane axis, $\boldsymbol{\Omega}_k = (0, 0, \Omega_k)^T$. In the planar extensional regime ($\chi = +1$), top-left (TL) and bottom-right (BR) cylinders rotate clockwise ($\Omega_k = -\Omega$), while top-right (TR) and bottom-left (BL) cylinders rotate counter-clockwise ($\Omega_k = +\Omega$), with peripheral angular speed $\Omega = u_{\text{roll}}/R$.

2. **No-Slip on Walls (Γ_{outer})**: Along the stationary external boundaries of the square cavity, the velocity vanishes identically:

$$\boldsymbol{u}|_{\Gamma_{\text{outer}}} = \mathbf{0}. \tag{2.37}$$

3. **Pressure Gauge Fixing**: Because pressure appears in the momentum balance (2.30) exclusively through its spatial gradient ∇p, the pressure field is determined only up to an arbitrary additive constant ($p \to p + C$) under pure velocity Dirichlet conditions. To eliminate this degree of freedom, pressure is anchored with a single reference point:

$$p(\boldsymbol{x}_0) = 0. \tag{2.38}$$

Chapter 3

Physics-Informed Neural Networks in Complex Rheology

Feedforward deep neural networks, commonly known as Multi-Layer Perceptrons (MLPs), provide flexible continuous approximations for non-linear functions [15]. An MLP is formed by composing a sequence of parametric functions, called *layers*. Each layer maps its inputs through a linear transformation followed by a non-linear operation. With these layers in series, the network can represent continuous functional mappings of arbitrary complexity without requiring explicit basis functions.

In scientific computing, neural networks represent physical fields directly as continuous functions over spatial coordinates, rather than storing discrete state variables on a mesh. However, using purely data-driven networks for fluid mechanics presents clear limits: they require large amounts of labeled data, can violate conservation laws, and cannot infer latent material properties on their own.

Physics-Informed Neural Networks (PINNs) address these limits by embedding governing partial differential equations directly into the training loss. The differential system formulated in Chapter 2, governed by mass conservation, vis-

coelastic momentum balance, and frame-indifferent constitutive transport, represents a coupled set of non-linear equations that poses well-known challenges to traditional mesh-based methods. In this chapter, we formulate this continuous boundary-value problem within a Scientific Machine Learning (SciML) framework. By embedding momentum and constitutive balances into the optimization objective through automatic differentiation, the PINN framework enables both forward state simulation and inverse rheological parameter discovery without spatial discretization meshes.

3.1 Fundamentals of Deep Learning

Deep learning models approximate continuous mathematical functions by learning parameters directly from data [15]. In this section, we review the fundamental building blocks of feedforward networks: the artificial neuron, activation functions, multi-layer architectures, loss functions, optimization algorithms, and gradient computation.

3.1.1 The Artificial Neuron and Non-Linear Activation Functions

The elementary computational unit of a neural network is the artificial neuron. Given an input vector $\boldsymbol{x}$, a neuron computes a linear combination of the inputs followed by a non-linear activation:

$$z = \boldsymbol{w} \cdot \boldsymbol{x} + b, \qquad a = \sigma(z), \tag{3.1}$$

where $\boldsymbol{w}$ is the weight vector, b is a scalar bias, $\sigma(\cdot)$ is the activation function and a is the output. Non-linear activations are essential: without them, composing multiple linear layers simply collapses into a single linear transformation.

In classical machine learning, piecewise-linear functions such as the Rectified Linear Unit, $\mathrm{ReLU}(z) = \max(0, z)$, are widely favored for their computational efficiency. However, in scientific computing and fluid mechanics, the network outputs parameterize continuous physical fields governed by differential equations. These governing balance laws involve spatial derivatives of first, second, or third order, such as viscous diffusion $\mu_s \nabla^2 \boldsymbol{u}$ or stress divergence $\nabla \cdot \boldsymbol{\tau}$.

Because the first derivative of ReLU is discontinuous and its second derivative is zero almost everywhere, evaluating a ReLU network inside differential residuals eliminates diffusive and curvature terms. Physics-informed models therefore require smooth, differentiable activation functions ($\mathcal{C}^2$ or higher). In this work, we employ the Sigmoid Linear Unit (SiLU, also known as Swish) [15]:

$$\mathrm{SiLU}(z) = \frac{z}{1 + e^{-z}}. \tag{3.2}$$

The SiLU activation is smooth and non-monotonic. Unlike the hyperbolic tangent (tanh), which saturates toward constant values for large positive inputs, SiLU remains active in the positive regime. This property maintains healthy gradient propagation in flow regions with intense shear and steep localized stress gradients, which are characteristic of viscoelastic flows.

3.1.2 Multi-Layer Perceptron Architecture

By arranging artificial neurons into sequential layers, we construct a Multi-Layer Perceptron (MLP). The network maps an input vector $\boldsymbol{x}$ to an output prediction

$\hat{\boldsymbol{u}}$ by propagating information through L successive layers [15]:

$$\begin{aligned} \boldsymbol{h}^{(1)} &= \sigma\left(\boldsymbol{W}^{(1)}\boldsymbol{x} + \boldsymbol{b}^{(1)}\right), \\ \boldsymbol{h}^{(l)} &= \sigma\left(\boldsymbol{W}^{(l)}\boldsymbol{h}^{(l-1)} + \boldsymbol{b}^{(l)}\right), \quad \text{for } l = 2, \ldots, L-1, \\ \hat{\boldsymbol{u}} &= \boldsymbol{W}^{(L)}\boldsymbol{h}^{(L-1)} + \boldsymbol{b}^{(L)}, \end{aligned} \tag{3.3}$$

where $\boldsymbol{h}^{(l)}$ denotes the hidden feature vector at layer l, and $\boldsymbol{W}^{(l)}$ and $\boldsymbol{b}^{(l)}$ represent the weight matrix and bias vector of that layer. In our implementation, the inputs $\boldsymbol{x} = (x, y)$ are spatial coordinates within the fluid cavity, while the outputs $\hat{\boldsymbol{u}}$ parameterize the physical state variables: the stream function ψ, hydrodynamic pressure p, and extra-stress tensor components $\boldsymbol{\tau}$.

A central theoretical property of MLPs is universal function approximation [14,23]: a feedforward network with sufficient width and smooth non-linear activations can approximate any continuous function on a compact domain to arbitrary precision. However, universal approximation guarantees only the existence of a suitable set of weights, not that standard gradient-based optimization will automatically find them or that spatial derivatives will match the physical solution with equal fidelity.

Importance of Weight Initialization

The initialization of network parameters strongly conditions the resulting optimization dynamics. Because deep architectures compose multiple transformations and non-linearities, inappropriate initial weight magnitudes can cause layer activations and backpropagated gradients to either vanish or explode across depth [15]. In scientific computing and differential problems, where non-linear convective terms and tensor couplings are evaluated, uncalibrated initial weights can trigger extreme initial residual spikes that immediately destabilize gradient

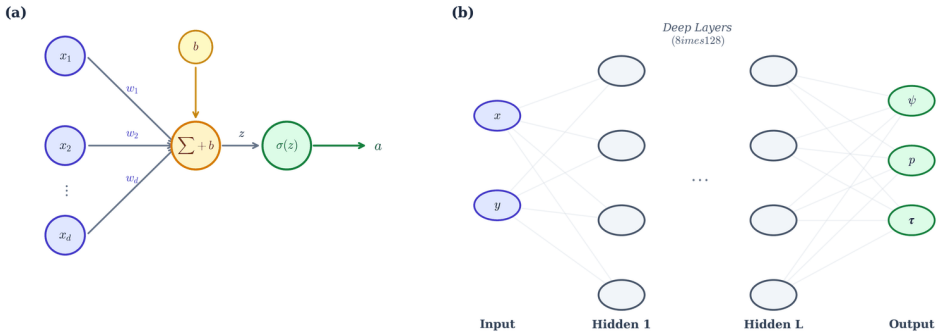


Figure 3.1: Neural network representations: (a) schematic of an artificial neuron computing an affine combination $z = \boldsymbol{w} \cdot \boldsymbol{x} + b$ followed by a smooth activation $\sigma(z)$; (b) deep Multi-Layer Perceptron (MLP) architecture propagating coordinate inputs (x, y) through L hidden layers to parameterize continuous physical fields $(\psi, p, \boldsymbol{\tau})$.

descent. Appropriate variance-preserving initialization schemes, such as scaling parameter distributions inversely with layer widths, help preserve stable gradient flow and avoid optimization divergence. Specific configuration choices regarding hidden and output layer initialization adopted in this work are detailed in Chapter 4.

3.1.3 Loss Functions and Optimization Algorithms

Training an MLP consists of identifying a parameter set $\boldsymbol{\theta} = \{\boldsymbol{W}^{(l)}, \boldsymbol{b}^{(l)}\}$ that minimizes a scalar objective function $\mathcal{L}(\boldsymbol{\theta})$. In supervised regression, the discrepancy between model predictions and target reference measurements is commonly quantified through the Mean Squared Error (MSE):

$$\mathcal{L}_{\text{MSE}}(\boldsymbol{\theta}) = \frac{1}{N} \sum_{i=1}^{N} \left\| \hat{\boldsymbol{u}}(\boldsymbol{x}_i; \boldsymbol{\theta}) - \boldsymbol{u}_i^{\text{target}} \right\|_2^2 . \tag{3.4}$$

Parameters are iteratively updated along descent directions derived from the loss gradient:

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta \boldsymbol{d}_k, \tag{3.5}$$

where $\eta > 0$ represents the learning rate (step size) and $\boldsymbol{d}_k$ denotes the search direction at iteration k.

In scientific machine learning, gradient-based solvers must navigate non-convex loss surfaces characterized by ill-conditioned curvature and local minima. Two widely adopted algorithms in this setting are:

1. **First-order adaptive optimization (Adam)**: The Adaptive Moment Estimation algorithm [22] computes individual adaptive learning rates for each parameter by maintaining exponentially decaying running averages of past gradients (first moment) and squared gradients (uncentered second moment). This moment-based scaling renders Adam particularly robust against ill-conditioned curvature and saddle points, making it highly effective for early-stage exploration across rugged non-convex landscapes.

2. **Second-order quasi-Newton optimization (L-BFGS)**: The Limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm [25] incorporates local curvature information by constructing low-rank approximations of the inverse Hessian matrix using the history of recent gradient evaluations. While higher-order computation requires evaluation over full data batches, L-BFGS provides deterministic superlinear convergence in the vicinity of local minima, capable of driving differential equation residuals to strict numerical tolerances.

Both algorithms share the common objective of driving the objective function toward an optimal stationary point, yet they exhibit distinct trade-offs: Adam offers robust stochastic exploration with modest per-iteration memory requirements, whereas L-BFGS delivers deterministic precision and rapid asymptotic convergence at the expense of curvature memory buffers.

3.1.4 Backpropagation

To update network parameters via gradient descent, we compute the partial derivatives of the scalar loss with respect to all weights and biases, $\nabla_{\boldsymbol{\theta}}\mathcal{L}$. Evaluating these gradients through individual numerical perturbations would be computationally prohibitive.

Instead, neural networks compute analytical gradients using the *Automatic Differentiation (AD)* algorithm [16]. Automatic differentiation evaluates derivatives to machine precision by applying the chain rule to elementary operations along the computational graph. In standard machine learning, reverse-mode AD is widely known as the *backpropagation* algorithm, primarily employed to evaluate the gradient of the scalar loss with respect to network parameters.

This reverse-mode automatic differentiation evaluates all parameter gradients simultaneously at a computational cost comparable to a single forward pass, making deep network training feasible.

3.2 Neural Networks: Capabilities and Physical Limitations

3.2.1 Key Advantages

Neural networks offer distinctive characteristics that make them attractive for continuous physical modeling:

1. **High-Dimensional Functional Approximation**: The universal approximation capacity allows neural networks to represent complex, coupled physical fields (velocities, pressures, and stress tensors) within a unified functional representation, without requiring specialized interpolation elements for each variable.

2. **Meshless Representation**: Traditional grid-based methods (Finite Element, Finite Volume, and Finite Difference Methods) represent state variables exclusively at discrete vertices or cell centers, relying on explicit mesh connectivity. In contrast, a neural network provides a continuous, globally defined mapping $\boldsymbol{x} \mapsto \hat{\boldsymbol{u}}(\boldsymbol{x})$ over the entire domain, capable of being evaluated at arbitrary spatial coordinates. However, being meshless does not imply freedom from geometric considerations: defining the domain, enforcing boundary conditions, and designing effective collocation point distributions remain essential and influential steps.

3. **Hardware Acceleration and Differentiability**: Modern deep learning frameworks are built upon dense linear algebra operations optimized for parallel accelerator hardware (GPUs and TPUs), and provide built-in automatic differentiation for computing spatial derivatives of the represented fields.

3.2.2 Limits of Purely Data-Driven Machine Learning

Despite their high expressivity, deploying purely data-driven black-box networks to fluid mechanics and non-Newtonian rheology reveals fundamental limitations:

- **Data scarcity in experimental rheology**: Deep neural networks typically rely on large labeled datasets. In fluid experiments, acquiring dense velocity fields via Particle Image Velocimetry (PIV) is labor-intensive, while non-intrusive measurements of internal extra-stress components are very hard to obtain in complex geometries.

- **Absence of physical grounding**: Purely data-driven models minimize empirical loss on discrete observations without physical constraints. Consequently, their predictions often violate incompressibility ($\nabla \cdot \boldsymbol{u} \neq 0$), mo-

mentum balance, and constitutive laws, generating non-physical artifacts when interpolating or extrapolating.

- **Overfitting**: In sparse-data regimes, over-parameterized networks can easily fit measurement noise, obtaining low training errors while losing accuracy in unmonitored regions.

- **Inability to discover governing parameters**: Black-box models learn input-output mappings, but cannot extract underlying physical properties, such as relaxation times or polymeric viscosities, that govern flow dynamics.

These limitations show that data alone are insufficient for scientific modeling in rheology, motivating frameworks that embed physical conservation laws directly into the loss function.

3.3 Physics-Informed Neural Network

The Physics-Informed Neural Network (PINN) framework, formalized by Raissi, Perdikaris, and Karniadakis [8], addresses the problems of purely data-driven modeling by incorporating differential partial equations directly into the network training objective.

3.3.1 Physics-Constrained Loss Formulation

Following the unified boundary-value problem (BVP) notation established in the scientific machine learning literature [24], consider a physical state described by a vector of continuous fields $\boldsymbol{u}(\boldsymbol{x})$ defined over a spatial domain $\Omega \subset \mathbb{R}^d$ bounded

by $\partial\Omega$:

$$\mathcal{F}[\boldsymbol{u}(\boldsymbol{x}); \boldsymbol{\lambda}] = \mathbf{0}, \quad \boldsymbol{x} \in \Omega, \tag{3.6}$$

$$\mathcal{B}[\boldsymbol{u}(\boldsymbol{x})] = \boldsymbol{g}(\boldsymbol{x}), \quad \boldsymbol{x} \in \partial\Omega, \tag{3.7}$$

where:

- $\boldsymbol{u}(\boldsymbol{x})$ denotes the physical state variables (in our viscoelastic framework, $\boldsymbol{u} = (\psi, p, \boldsymbol{\tau})$, comprising stream function, pressure, and extra-stress tensor components);
- $\mathcal{F}$ is a general non-linear differential operator encoding the governing field equations (mass conservation, momentum balance, and constitutive transport equations);
- $\boldsymbol{\lambda}$ denotes physical or constitutive parameters governing the flow dynamics (such as solvent viscosity, polymeric viscosity, and relaxation time);
- $\mathcal{B}$ denotes the boundary operator enforcing kinematic constraints on the boundaries $\partial\Omega$ (such as no-slip conditions along solid walls and prescribed angular velocities on roller surfaces), with prescribed boundary values $\boldsymbol{g}(\boldsymbol{x})$.

In a PINN, the continuous solution fields are parameterized by a deep neural network surrogate $\hat{\boldsymbol{u}}(\boldsymbol{x}; \boldsymbol{\theta})$ parameterized by weights and biases $\boldsymbol{\theta}$. Evaluating the governing differential operator on the network outputs via automatic differentiation defines the pointwise differential residual:

$$\boldsymbol{f}(\boldsymbol{x}; \boldsymbol{\theta}, \boldsymbol{\lambda}) \equiv \mathcal{F}[\hat{\boldsymbol{u}}(\boldsymbol{x}; \boldsymbol{\theta}); \boldsymbol{\lambda}]. \tag{3.8}$$

The network parameters $\boldsymbol{\theta}$, alongside any unknown physical parameters $\boldsymbol{\lambda}$ in inverse problems, are optimized simultaneously by minimizing a composite multi-

objective loss:

$$\mathcal{L}_{\text{total}}(\boldsymbol{\theta}, \boldsymbol{\lambda}) = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{pde}}\mathcal{L}_{\text{pde}}, \tag{3.9}$$

where each individual loss component is evaluated as:

$$\mathcal{L}_{\text{data}} = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} \left\| \hat{\boldsymbol{u}}(\boldsymbol{x}_i^{\text{data}}) - \boldsymbol{u}_i^{\text{data}} \right\|_2^2, \tag{3.10}$$

$$\mathcal{L}_{\text{bc}} = \frac{1}{N_{\text{bc}}} \sum_{j=1}^{N_{\text{bc}}} \left\| \mathcal{B}[\hat{\boldsymbol{u}}(\boldsymbol{x}_j^{\text{bc}})] - \boldsymbol{g}(\boldsymbol{x}_j^{\text{bc}}) \right\|_2^2, \tag{3.11}$$

$$\mathcal{L}_{\text{pde}} = \frac{1}{N_{\text{pde}}} \sum_{k=1}^{N_{\text{pde}}} \left\| \boldsymbol{f}(\boldsymbol{x}_k^{\text{pde}}; \boldsymbol{\theta}, \boldsymbol{\lambda}) \right\|_2^2. \tag{3.12}$$

Here, $\{\boldsymbol{x}_i^{\text{data}}\}$ denotes available empirical observations or measurement locations, $\{\boldsymbol{x}_j^{\text{bc}}\}$ represents points sampled along physical boundaries $\partial\Omega$, and $\{\boldsymbol{x}_k^{\text{pde}}\}$ designates interior *collocation points* scattered across Ω.

This formulation naturally supports two complementary problem settings:

- **Forward problems**: The physical parameters $\boldsymbol{\lambda}$ are known. The network optimizes its weights $\boldsymbol{\theta}$ to solve for the continuous fields $\hat{\boldsymbol{u}}(\boldsymbol{x})$, acting as a meshless continuous PDE solver.

- **Inverse problems**: One or more constitutive parameters in $\boldsymbol{\lambda}$ are unknown. The network simultaneously reconstructs the unobserved latent fields and identifies the unknown parameters by combining empirical velocity observations $\mathcal{L}_{\text{data}}$ with physical consistency constraints $\mathcal{L}_{\text{pde}}$.

A key methodological principle of this thesis is the distinction between field reconstruction accuracy and parameter identifiability. A network may fit observed velocity fields accurately, yet inferred rheological parameters may remain uncertain if the kinematics lack sufficient variation or if the loss landscape exhibits degeneracies. Embedding physical equations provides essential structure,

but does not automatically guarantee unique parameter recovery.

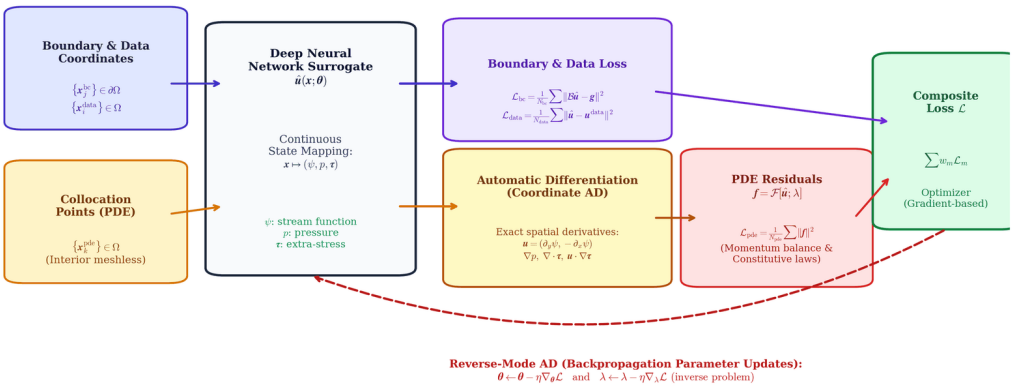


Figure 3.2: Schematic computational workflow of a Physics-Informed Neural Network (PINN): boundary coordinates $\{\boldsymbol{x}_j^{\text{bc}}\}$ and measurements $\{\boldsymbol{x}_i^{\text{data}}\}$ evaluate boundary adherence ($\mathcal{L}_{\text{bc}}$) and empirical matching ($\mathcal{L}_{\text{data}}$) (top branch), while interior collocation points $\{\boldsymbol{x}_k^{\text{pde}}\}$ pass through Coordinate Automatic Differentiation (AD) to compute exact spatial derivatives and differential PDE residuals ($\mathcal{L}_{\text{pde}}$, bottom branch). The composite loss $\mathcal{L}_{\text{total}}$ drives gradient descent via reverse-mode backpropagation, jointly optimizing surrogate weights $\boldsymbol{\theta}$ and identifying unknown physical parameters $\boldsymbol{\lambda}$ in inverse settings.

3.3.2 PINN-Specific Features

Several operational features distinguish physics-informed training from conventional supervised learning:

1. **Training as a boundary-value solver**: In classical machine learning, a model is trained on a sample set to generalize to unseen statistical distributions. In PINNs, *training is the numerical solver*. The network is not designed to predict flows across arbitrary unseen geometries; rather, optimization directly solves the specific problem on the specified domain Ω. Available measurements $\mathcal{L}_{\text{data}}$ act as an active data anchor that guides the network toward the true physical state.

2. **Physical regularization and dynamic loss weighting**: In standard

supervised training, the network is unconstrained between data points. In PINNs, evaluating differential residuals $\mathcal{L}_{\text{pde}}$ across interior collocation points penalizes non-physical gradients and curvatures, restricting the admissible solution space. However, balancing the competing loss weights (w_{data}, w_{bc}, and w_{pde}) presents practical challenges. As analyzed by Wang, Teng, and Perdikaris [12], the backpropagated gradient magnitudes arising from PDE residuals and boundary constraints often differ by several orders of magnitude. This imbalance induces numerical stiffness, where boundary terms can dominate and suppress the learning signal of interior PDE residuals (or vice versa). Calibrated weighting between data, boundary, and differential components is therefore needed to prevent optimization stalls and ensure stable convergence.

3. **Coordinate Automatic Differentiation (AD)**: Automatic differentiation serves a dual purpose in PINNs: beyond computing loss gradients with respect to network parameters, it evaluates exact spatial derivatives of the predicted physical fields with respect to coordinates:

$$\text{Coordinate AD (PINNs):} \quad \frac{\partial \hat{\boldsymbol{u}}}{\partial \boldsymbol{x}}, \quad \frac{\partial^2 \hat{\boldsymbol{u}}}{\partial \boldsymbol{x}^2}. \tag{3.13}$$

The accuracy of these derivatives with respect to the true physical field depends on how closely the network itself approximates the physical solution.

4. **Collocation point sampling**: Unlike traditional CFD solvers that require meshes and element connectivity tables, PINNs evaluate differential residuals at sparse collocation points $\{\boldsymbol{x}_k^{\text{pde}}\}$. This meshless feature simplifies domain representation, though sufficient point density remains necessary in boundary layers and near stagnation regions.

Table 3.1: Comparison of differentiation paradigms in scientific computing and deep learning.

Method	Formulation	Truncation Error	Grid Dependence
Finite Differences (FDM)	Discrete difference quotients (Taylor series)	$\mathcal{O}(\Delta x^2)$	Structured Mesh Stencil
Symbolic Differentiation	Closed-form analytical expression	None	Expression Growth
Automatic Differentiation (AD)	Exact chain rule on computational graph	**None (Machine Precision)**	**Meshless / Continuous**

3.4 Foundational Literature Benchmarks

3.4.1 Navier–Stokes Flow Past a Cylinder (Raissi et al., 2019)

In their foundational paper, Raissi, Perdikaris, and Karniadakis [8] demonstrated the capability of PINNs on laminar vortex past a cylinder at $Re = 100$.

Problem Formulation

The dimensionless incompressible Navier-Stokes equations were formulated as:

$$u_t + \lambda_1(uu_x + vu_y) = -p_x + \lambda_2(u_{xx} + u_{yy}), \tag{3.14}$$

$$v_t + \lambda_1(uv_x + vv_y) = -p_y + \lambda_2(v_{xx} + v_{yy}), \tag{3.15}$$

$$u_x + v_y = 0, \tag{3.16}$$

where $\lambda_1 = 1.0$ governs convective acceleration and $\lambda_2 = 1/Re = 0.01$ represents kinematic viscosity.

To satisfy incompressibility identically without requiring an explicit divergence penalty, the authors adopted the stream function formulation established in Section 2.2.1 (Eq. 2.10), parameterizing velocity components through spatial derivatives of $\psi(t, x, y)$. A neural network with 9 hidden layers and 20 neurons per layer, activated by tanh, predicted (ψ, p) while treating λ_1 and λ_2 as learnable parameters.

Field and Parameter Identification

The network was trained on $N = 5{,}000$ spatio-temporal velocity points. The benchmark demonstrated two key capabilities:

- **Pressure recovery**: Without any pressure measurements, the network reconstructed the pressure field $p(t, x, y)$ exclusively relying on momentum balance, matching CFD reference data up to an arbitrary constant.
- **Parameter estimation under noise**: On clean data, parameter errors were 0.08% for λ_1 and 4.67% for λ_2. Under 1% uncorrelated Gaussian noise, errors remained controlled at 0.17% and 5.70%.

While this benchmark validated simultaneous field reconstruction and parameter identification, it was restricted to linear Newtonian fluids in an open domain.

3.4.2 The ViscoelasticNet Framework (Thakur et al., 2024)

Extending physics-informed methods to non-Newtonian fluids, Thakur, Raissi, and Ardekani [9] introduced *ViscoelasticNet*, applying PINNs to polymeric flows governed by differential constitutive equations.

The Rheological Bottleneck

Viscoelastic fluid simulation presents two major challenges:

1. **Experimental limitations**: Non-intrusive measurements of internal extra-stresses $\boldsymbol{\tau}$ are generally unavailable in complex geometries.
2. **Numerical instability (HWNP)**: In standard grid-based solvers (FEM, FVM), elevated elasticity triggers the High-Weissenberg Number Problem (HWNP), where steep stress boundary layers cause convergence breakdown unless stabilized numerically.

Virtual Rheometer

To address these limits, ViscoelasticNet proposed an inverse framework operating as a "virtual rheometer":

- **Stress inference**: Given velocity measurements $\boldsymbol{u}$, the network infers unobserved extra-stresses $\boldsymbol{\tau}$ and pressure p by enforcing constitutive and momentum residuals.
- **Unified constitutive formulation**: The authors formulated a unified constitutive expression encompassing Oldroyd-B, linear PTT, and Giesekus models:
$$\boldsymbol{\tau} + \lambda \overset{\nabla}{\boldsymbol{\tau}} + \frac{\alpha\lambda}{\mu_p}(\boldsymbol{\tau}\cdot\boldsymbol{\tau}) + \frac{\varepsilon\lambda}{\mu_p}\mathrm{tr}(\boldsymbol{\tau})\boldsymbol{\tau} = 2\mu_p \boldsymbol{D}, \tag{3.17}$$
seeking to identify λ, μ_p, α, and ε.
- **Decoupled two-stage training**: Optimization was separated into two sequential stages:
  - *Stage 1 (Stress and Rheology)*: Velocity and stress networks were optimized jointly using velocity data and constitutive residuals, learning rheological parameters $(\lambda, \mu_p, \alpha, \varepsilon)$.
  - *Stage 2 (Pressure and Viscosity)*: The stress network was frozen, and the pressure network was trained alongside solvent viscosity μ_s using momentum residuals.
- **Investigated geometries**: ViscoelasticNet was evaluated on two open geometries: a planar stenosed channel and a cross-slot flow. The authors reported relative stress errors below 5% and demonstrated stability under synthetic velocity noise up to 10%.

These results confirmed that PINNs can infer non-Newtonian stresses from velocity fields in open channel flows. However, their formulation focused on do-

mains with net through-flow and convective pressure drops. In the next chapter, we address the open problem of enclosed recirculating flows: we formulate a stream-function PINN for the four-roll mill geometry, evaluate forward field accuracy, and investigate the boundaries of inverse parameter discovery.

Chapter 4

Results and Discussion: Four-Roll Mill PINN Solver

This chapter validates the viscoelastic PINN framework on the four-roll mill benchmark and then deploys it as an inverse virtual rheometer. We first establish the problem setting, network architecture, and two-stage training strategy; we then verify the forward solver against high-fidelity COMSOL reference data. The core of the chapter applies the inverse solver to Oldroyd-B, Giesekus, and PTT fluids, exposing both the strengths and the physical identifiability limits of the approach.

4.1 Problem Statement and Novel Contributions

4.1.1 Challenges of Viscoelastic PINNs in Recirculating Cavities

In complex polymeric fluid flows, conventional computational fluid dynamics (CFD) methods based on Finite Element (FEM) or Finite Volume (FVM) discretizations face two fundamental bottlenecks:

1. **The High-Weissenberg Number Problem (HWNP)**: Hyperbolic convective stress transport generates steep localized stress boundary layers and thin birefringent strands near solid boundaries and stagnation points. In traditional mesh-based methods, these steep gradients trigger numerical instabilities, spatial oscillations, and convergence breakdown unless specialized stabilization techniques (such as DEVSS or log-conformation formulations) are incorporated [3].

2. **Inverse Parameter Identification Complexity**: Standard CFD solvers are inherently forward predictive engines. Inferring unknown rheological parameters (e.g., relaxation times or polymeric viscosities) from experimental kinematic velocity measurements requires nested, computationally prohibitive PDE-constrained optimization loops.

Physics-Informed Neural Networks [8,24] reformulate both simulation and parameter discovery as a continuous, meshless optimization problem. While prior work such as ViscoelasticNet [9] demonstrated the promise of PINNs for viscoelastic stress estimation, the literature leaves critical challenges unresolved:

- **Enclosed Recirculating Cavities with Mixed Kinematics**: Prior viscoelastic PINN benchmarks have focused almost exclusively on open through-flow channels (such as planar contractions, stenosed channels, or cross-slots) driven by net convective pressure gradients. In contrast, the four-roll mill benchmark investigated here is a closed square cavity featuring recirculating streamlines with zero net mass flux, driven entirely by rotating cylindrical rollers [18]. This setup combines a central hyperbolic extensional stagnation point with intense simple shear in the roller gaps. Capturing closed stress orbits without unphysical spatial accumulation or numerical dissipation represents a severe test of PINN stability.

- **Parameter Identifiability in Extended Rheological Regimes**: Existing viscoelastic PINN studies have typically explored restricted parameter domains (relaxation times $\lambda \leq 0.15\,\mathrm{s}$, polymeric viscosities $\mu_p \leq 0.025\,\mathrm{Pa}\cdot\mathrm{s}$, and mobility $\alpha \leq 0.20$). Extending investigations to substantially larger relaxation times ($\lambda \leq 1.0\,\mathrm{s}$) and higher viscosities brings the system into regimes where non-linear convective stress terms interact strongly with viscous dissipation, testing the limits of parameter identifiability [19].

4.1.2 Forward and Inverse Operational Modes

The mathematical formulation naturally accommodates two distinct operational settings:

- **Forward (Direct) Problem**: Given fully specified fluid properties (λ, μ_p, μ_s, and non-linear parameters α or ε) alongside kinematic boundary conditions on solid walls and rollers, the network optimizes its surrogate parameters from scratch to predict the continuous velocity field $\boldsymbol{u}(x, y)$, pressure field $p(x, y)$, and polymeric extra-stress tensor $\boldsymbol{\tau}(x, y)$ that satisfy momentum balance and constitutive closure with zero interior supervision.

- **Inverse (Virtual Rheometer) Problem**: Given kinematic observations of the velocity field within the cavity (emulating planar Particle Image Velocimetry, PIV), the network reconstructs the unobserved latent fields $(p, \boldsymbol{\tau})$ while simultaneously identifying the unknown governing physical parameters $(\lambda, \mu_p, \mu_s, \alpha, \varepsilon)$ through physics-constrained loss minimization.

Table 4.1 summarizes the operational role of each field variable and physical parameter across the two problem formulations.

Table 4.1: Classification of physical fields and parameters across forward and inverse problem formulations.

Physical Quantity	**Symbol**	**Forward Problem**	**Inverse Problem**
Velocity Field	$\boldsymbol{u} = (u, v)$	Primary unknown (via ψ)	Observed
Pressure Field	p	Primary unknown	Latent (unobserved)
Polymeric Extra-Stress	$\boldsymbol{\tau} = (\tau_{xx}, \tau_{xy}, \tau_{yy})$	Primary unknown	Latent (unobserved)
Relaxation Time	λ (Wi)	Known parameter	Unknown parameter
Polymeric Viscosity	μ_p	Known parameter	Unknown parameter
Solvent Viscosity	μ_s	Known parameter	Unknown parameter
Giesekus Mobility	α	Known parameter	Unknown parameter
PTT Extensibility	ε	Known parameter	Unknown parameter

4.2 Computational Architecture and Two-Stage Methodology

4.2.1 Decoupled Three-Head Stream-Function Network

To satisfy mass conservation identically in two-dimensional steady incompressible flow ($\nabla \cdot \boldsymbol{u} = 0$), the velocity field is parameterized via a scalar stream function $\psi(x, y)$ according to Equation (2.10):

$$u = \frac{\partial \psi}{\partial y}, \qquad v = -\frac{\partial \psi}{\partial x}. \tag{4.1}$$

Because mixed second partial derivatives commute by Schwarz's theorem, $\nabla \cdot \boldsymbol{u} = \frac{\partial^2 \psi}{\partial x \partial y} - \frac{\partial^2 \psi}{\partial y \partial x} \equiv 0$ holds identically everywhere in the computational domain. This eliminates the incompressibility constraint from the loss function, removing penalty weight tuning for continuity.

As depicted in Figure 4.1, the physical state is predicted by three independent Multi-Layer Perceptron (MLP) subnets rather than a single monolithic output

network:

$$(x,y) \in \mathbb{R}^2 \longrightarrow \begin{cases} \psi(x,y) \in \mathbb{R}^1 & \text{(Stream function, parameterizing } u, v \text{ via AD)}, \\ p(x,y) \in \mathbb{R}^1 & \text{(Hydrodynamic pressure)}, \\ \boldsymbol{\tau}(x,y) = (\tau_{xx}, \tau_{xy}, \tau_{yy}) \in \mathbb{R}^3 & \text{(Symmetric extra-stress tensor)}. \end{cases} \tag{4.2}$$

Each individual subnet consists of 8 hidden layers with 128 neurons per layer, activated by the smooth, infinitely differentiable Sigmoid Linear Unit (SiLU$(z) = z/(1+e^{-z})$). Smooth non-monotonic activations prevent vanishing second- and third-order derivative graphs during Coordinate Automatic Differentiation.

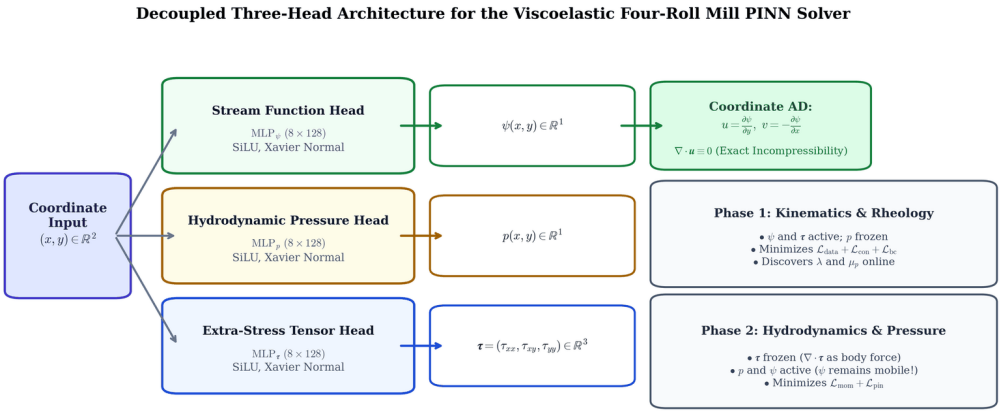


Figure 4.1: The decoupled three-head neural network architecture for the viscoelastic four-roll mill solver, showing coordinate inputs, separate MLP heads for stream function ψ, pressure p, and stress tensor $\boldsymbol{\tau}$, coordinate automatic differentiation, and the two-stage optimization schedule.

Physics-Informed Weight Initialization and Output Zero-Init

Hidden layers are initialized with the Xavier/Glorot normal distribution, scaling parameter variance inversely with layer widths to ensure well-conditioned gradient flow across depth.

The output layers of the extra-stress network $\mathcal{N}_{\boldsymbol{\tau}}$ and pressure network $\mathcal{N}_p$ are initialized to zero ($\boldsymbol{W}^{(L)} = \mathbf{0}, \boldsymbol{b}^{(L)} = \mathbf{0}$). This choice has two practical benefits:

1. **Physical state of rest**: At epoch zero, the model starts from zero extra-stress ($\boldsymbol{\tau} = \mathbf{0}$) and uniform pressure, reflecting a fluid at rest before shearing begins.

2. **Initial gradient stability**: Random output weights generate high-frequency oscillations in space. Because viscoelastic equations include quadratic terms ($\alpha\boldsymbol{\tau}^2$ in Giesekus) and convective derivatives ($\boldsymbol{u} \cdot \nabla\boldsymbol{\tau}$), evaluating uncalibrated stress fields at iteration zero can produce loss spikes ($> 10^6$) that disrupt optimizer momentum. Zero-initializing the outputs prevents these initial spikes, allowing residuals to grow smoothly as kinematics are resolved.

4.2.2 Multi-Objective Loss, Explicit Residuals, and Stress Scaling Anchor

The composite objective function minimized during network training is formulated as:

$$\mathcal{L}_{\text{total}} = w_{\text{data}}\mathcal{L}_{\text{data}} + w_{\text{bc}}\mathcal{L}_{\text{bc}} + w_{\text{con}}\mathcal{L}_{\text{con}} + w_{\text{mom}}\mathcal{L}_{\text{mom}} + w_{\text{roll_stress}}\mathcal{L}_{\text{roll_stress}} + w_{\text{pin}}\mathcal{L}_{\text{pin}}. \tag{4.3}$$

Dimensional-to-Dimensionless Scaling Alignment

In Chapter 2 (Section 2.4), non-dimensional equations were derived using reference length $H_{\text{ref}} = R = 0.005\,\text{m}$ (roller radius), reference speed $U_{\text{ref}} = u_{\text{roll}} = 8.333 \times 10^{-3}\,\text{m/s}$, and characteristic viscosity $\eta_0 = \mu_0 = \mu_s + \mu_p$. Under this convention, the dimensionless viscosities equal the viscosity ratios: $\mu_s^* = \beta$ and $\mu_p^* = 1 - \beta$.

In the computational inverse framework, however, the fluid viscosities μ_s and μ_p are unknown a priori and must be discovered by the network. Therefore,

the computational solver adopts a fixed, constant viscosity scale $\eta_0 = 1.0\,\mathrm{Pa}\cdot\mathrm{s}$, decoupling the scaling from the evolving parameter estimates:

$$\mu_s^* = \frac{\mu_s}{\eta_0}, \quad \mu_p^* = \frac{\mu_p}{\eta_0}, \quad Re_{\text{scale}} = \frac{\rho U_{\text{ref}} H_{\text{ref}}}{\eta_0}, \quad Wi = \frac{\lambda U_{\text{ref}}}{H_{\text{ref}}}. \tag{4.4}$$

When the true total viscosity equals η_0, $\mu_s^* = \beta$ and $\mu_p^* = 1 - \beta$ are recovered identically. Furthermore, while the physical domain $[-L/2, L/2]^2 = [-25, 25]\,\mathrm{mm}$ corresponds to $[-5, 5]^2$ when scaled by $H_{\text{ref}} = R$, network coordinate inputs are mapped to the unit box $[0, 1]^2$ using the cavity width $H_{\text{coord}} = L = 0.05\,\mathrm{m}$ to ensure well-conditioned activation arguments. The spatial gradient operator is appropriately rescaled via Coordinate AD by $(H_{\text{ref}}/H_{\text{coord}})$.

Explicit PDE Residuals

Evaluating the differential operators via automatic differentiation at interior collocation points $\boldsymbol{x}_m \in \Omega$ yields the explicit residuals:

1. **Linear Momentum Residuals** (f_u, f_v):

$$f_u = Re_{\text{scale}} \left(u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} \right) + \frac{\partial p}{\partial x} - \mu_s^* \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) - \left(\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} \right), \tag{4.5}$$

$$f_v = Re_{\text{scale}} \left(u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y} \right) + \frac{\partial p}{\partial y} - \mu_s^* \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) - \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} \right). \tag{4.6}$$

2. **Viscoelastic Constitutive Residuals** $(f_{\tau_{xx}}, f_{\tau_{xy}}, f_{\tau_{yy}})$:

$$f_{\tau_{xx}} = \mathcal{G}_{xx}(\boldsymbol{\tau}) + Wi\,\overset{\nabla}{\tau}_{xx} - 2\mu_p^* \frac{\partial u}{\partial x}, \tag{4.7}$$

$$f_{\tau_{yy}} = \mathcal{G}_{yy}(\boldsymbol{\tau}) + Wi\,\overset{\nabla}{\tau}_{yy} - 2\mu_p^* \frac{\partial v}{\partial y}, \tag{4.8}$$

$$f_{\tau_{xy}} = \mathcal{G}_{xy}(\boldsymbol{\tau}) + Wi\,\overset{\nabla}{\tau}_{xy} - \mu_p^* \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right), \tag{4.9}$$

where $\overset{\nabla}{\tau}_{ij}$ denotes the upper-convected time derivative (Eq. (2.17)), and $\boldsymbol{\mathcal{G}}(\boldsymbol{\tau})$ encapsulates the constitutive model:

$$\boldsymbol{\mathcal{G}}(\boldsymbol{\tau}) = \begin{cases} \boldsymbol{\tau} & \text{(Oldroyd-B, } \alpha = 0\text{)}, \\ \boldsymbol{\tau} + \frac{\alpha Wi}{\mu_p^*}\boldsymbol{\tau}^2 & \text{(Giesekus, } \alpha > 0\text{)}, \\ \left(1 + \frac{\varepsilon Wi}{\mu_p^*}\mathrm{tr}(\boldsymbol{\tau})\right)\boldsymbol{\tau} & \text{(Linear PTT, } \varepsilon > 0\text{)}. \end{cases} \tag{4.10}$$

Role of Roller Stress Supervision ($\mathcal{L}_{\text{roll_stress}}$)

The role of the roller stress term,

$$\mathcal{L}_{\text{roll_stress}} = \frac{1}{N_{\text{rolls}}} \sum_{k=1}^{N_{\text{rolls}}} \left\| \boldsymbol{\tau}_k - \boldsymbol{\tau}_k^{\text{ref}} \right\|_2^2, \tag{4.11}$$

differs fundamentally between the two problem settings:

- **Forward problem**: When fluid properties (λ, μ_p, μ_s) are known, the stress scale is determined by the $2\mu_p \boldsymbol{D}$ source term and the prescribed roller velocities. The network needs no stress supervision on the boundaries ($w_{\text{roll_stress}} = 0$) and solves the coupled PDEs autonomously.
- **Inverse problem**: In creeping flow ($Re = 0.0417 \ll 1$), inertial forces are negligible. Multiplying viscosities (μ_s, μ_p), pressure p, and stress $\boldsymbol{\tau}$ by an arbitrary constant $C > 0$ preserves the solenoidal velocity field $\boldsymbol{u}(x, y)$:

$$-\nabla(Cp) + (C\mu_s)\nabla^2\boldsymbol{u} + \nabla\cdot(C\boldsymbol{\tau}) = C\left[-\nabla p + \mu_s\nabla^2\boldsymbol{u} + \nabla\cdot\boldsymbol{\tau}\right] = \mathbf{0}. \tag{4.12}$$

Velocity measurements alone can therefore fix dimensionless groups (Wi, β), but not the absolute dimensional scale in Pa · s. Prescribing $\boldsymbol{\tau}$ on the roller boundaries provides the required dimensional traction anchor, resolving this scaling ambiguity without providing any interior stress data.

Table 4.2 summarizes the exact supervision data supplied across all investigated benchmarks.

Table 4.2: Supervision and boundary data supplied to the PINN framework across all experimental benchmarks.

Benchmark	**Domain Velocity** ($\mathcal{L}_{\text{data}}$)	**Wall/Roll Kinematic BCs** ($\mathcal{L}_{\text{bc}}$)	**Roll Stress Anchor** ($\mathcal{L}_{\text{roll_stress}}$)	**Pressure Pin** ($\mathcal{L}_{\text{pin}}$)	**Internal Stress** ($\boldsymbol{\tau}_{\text{interior}}$)
Direct Problem	None ($w_{\text{data}} = 0$)	Prescribed $\boldsymbol{u}$	**None** ($w_{\text{roll_stress}} = 0$, autonomous)	At $\boldsymbol{x}_0$ (Phase 2)	**None (Latent)**
Inverse (Cases 1–4)	Distributed (u, v)	Prescribed $\boldsymbol{u}$	Prescribed on rolls (Scale Anchor)	At $\boldsymbol{x}_0$ (Phase 2)	**None (Latent)**
Collocation Study	Distributed at N_{col}	Prescribed $\boldsymbol{u}$	Prescribed on rolls (Scale Anchor)	At $\boldsymbol{x}_0$ (Phase 2)	**None (Latent)**
PTT Degeneracy	Distributed (u, v)	Prescribed $\boldsymbol{u}$	Prescribed on rolls (Scale Anchor)	At $\boldsymbol{x}_0$ (Phase 2)	**None (Latent)**

4.2.3 Two-Stage Physical Decoupling and Helmholtz-Hodge Rationale

Simultaneous end-to-end optimization of coupled momentum and constitutive equations frequently suffers from spectral loss imbalance: backpropagated gradient norms from hyperbolic stress transport terms dominate those of elliptic momentum terms [9, 12]. We decouple the training trajectory into two distinct phases:

1. **Phase 1: Kinematics and Rheology (Stress Network Optimization)**: Momentum balance is disabled ($w_{\text{mom}} = 0$) and the pressure network $\mathcal{N}_p$ is frozen. Optimization trains the stream function ψ and stress tensor $\boldsymbol{\tau}$ jointly by minimizing kinematic data matching and constitutive residuals ($\mathcal{L}_{\text{con}} \to 0$). In inverse mode, unknown fluid parameters (λ, μ_p, α) are learned online, isolating hyperbolic stress dynamics from pressure gradient interference.

2. **Phase 2: Hydrodynamics and Pressure Recovery (Momentum Optimization)**: The extra-stress network $\mathcal{N}_{\boldsymbol{\tau}}$ is frozen, with its spatial divergence $\nabla \cdot \boldsymbol{\tau}$ acting as a fixed body force. The momentum loss is activated ($w_{\text{mom}} = 1.0$) alongside a pressure gauge anchor ($w_{\text{pin}} = 10.0$) at $\boldsymbol{x}_0$ to train the pressure network $\mathcal{N}_p$.

Mobile Stream Function in Phase 2 (Helmholtz-Hodge Rationale)

The stream function network $\mathcal{N}_\psi$ remains trainable under a reduced learning rate during Phase 2. The physical reason is tied to the Helmholtz-Hodge decomposition. Taking the curl of the momentum balance (Eq. (4.5)–(4.6)) eliminates the pressure gradient identically:

$$\nabla \times \nabla p \equiv \mathbf{0} \implies \nabla \times \left(\mu_s \nabla^2 \boldsymbol{u} + \nabla \cdot \boldsymbol{\tau} - \rho(\boldsymbol{u} \cdot \nabla \boldsymbol{u})\right) = \mathbf{0}. \tag{4.13}$$

Because pressure produces purely irrotational forces, ∇p cannot compensate any solenoidal (rotational) mismatch between viscous diffusion $\mu_s \nabla^2 \boldsymbol{u}$ and stress divergence $\nabla \cdot \boldsymbol{\tau}$. If ψ were frozen after Phase 1, even a small rotational imperfection in the velocity field would leave the momentum balance incompatible: no pressure field $p(x, y)$ could satisfy it. Keeping ψ mobile allows the solver to make minor solenoidal corrections that reconcile kinematics with momentum conservation.

Table 4.3 summarizes the training phases, trainable components, and loss weights.

Table 4.3: Summary of the decoupled two-stage training strategy for the four-roll mill PINN solver.

Training Phase	Trainable Heads	Frozen Heads	Active Losses	Key Loss Weights	Target Physics
Phase 1	ψ, $\boldsymbol{\tau}$	p	$\mathcal{L}_{\text{data}}, \mathcal{L}_{\text{bc}}, \mathcal{L}_{\text{con}}, \mathcal{L}_{\text{roll_stress}}$	$w_{\text{data}} = 1, w_{\text{con}} = 1, w_{\text{mom}} = 0$	λ, μ_p, α, Stress tensor
Phase 2	p, ψ (mobile)	$\boldsymbol{\tau}$	$\mathcal{L}_{\text{data}}, \mathcal{L}_{\text{bc}}, \mathcal{L}_{\text{mom}}, \mathcal{L}_{\text{pin}}$	$w_{\text{data}} = 20, w_{\text{mom}} = 1, w_{\text{pin}} = 10$	Pressure field p, Momentum balance

4.2.4 Dual-Precision Optimization Schedule and Transfer Learning Protocol

Staged Precision Strategy (Adam FP32 to L-BFGS FP64)

Optimization proceeds in two sequential precision tiers:

1. **Tier 1: Global Exploration via Adam in FP32**: During early training, the non-convex multi-objective landscape features steep valleys and local

saddles. The Adam optimizer operates in single precision (FP32), leveraging hardware tensor cores for fast throughput. A Cosine Annealing learning rate schedule ($2.5 \times 10^{-3} \rightarrow 2.5 \times 10^{-6}$) guides parameter trajectories into the basin of attraction of the physical ground state.

2. **Tier 2: Asymptotic Physical Refinement via L-BFGS in FP64**: To eliminate stochastic gradient noise and resolve higher-order spatial derivatives without round-off cancellation, the entire model is converted to double precision (`torch.float64`). Operating with a Strong Wolfe line search and history size $m = 50$, L-BFGS constructs low-rank inverse Hessian approximations, providing deterministic superlinear convergence that drives PDE residuals to scientific tolerances.

Transfer Learning (TL) Protocol

Training PINNs from scratch for every fluid regime requires substantial computational budgets (50,000 iterations) and risks local minima at high elasticity. We establish a Transfer Learning protocol:

1. Pre-train the base network on a moderate baseline case ($\lambda = 0.05\,\mathrm{s}, Wi = 0.08$).

2. Transfer pre-trained weights as initializations for target regimes characterized by high elasticity ($\lambda = 0.70\,\mathrm{s}, Wi = 1.17$) or strong non-linearity ($\alpha = 0.50$).

3. Fine-tune with halved iteration budgets (25,000 iterations), achieving a 50% reduction in computational duration while preventing high-Wi optimization divergence.

Table 4.4 details the hyperparameters and iteration budgets for both precision tiers and initialization strategies.

Table 4.4: Staged optimization hyperparameters, precision schedule, and transfer learning settings.

Tier	Algorithm	Precision	Iterations (Ex-Novo / TL)	Learning Rate (η) / Step Length	Line Search / Decay
Tier 1	Adam	FP32 (Single)	40,000 / 20,000	$2.5 \times 10^{-3} \rightarrow 2.5 \times 10^{-6}$	Cosine Annealing
Tier 2	L-BFGS	FP64 (Double)	10,000 / 5,000	$\alpha_0 = 1.0$ (unit initial step length)	Strong Wolfe line search

4.3 High-Fidelity Reference CFD and Mesh Convergence

4.3.1 COMSOL Finite-Element Setup and Worst-Case Benchmark

Reference solutions were generated using COMSOL Multiphysics, solving steady incompressible viscoelastic flows on unstructured triangular finite elements refined near roller surfaces.

To verify that reference fields are free from discretization artifacts, a mesh convergence study was performed on the Oldroyd-B benchmark at $\lambda = 1.0\,\mathrm{s}$ ($Wi = 1.667$). Linear Oldroyd-B lacks non-linear stress saturation, so extensional stresses grow most rapidly near the central stagnation point, making this the most demanding configuration for grid resolution. Establishing independence here gives confidence in the ground truth across all milder regimes.

We investigated six mesh resolutions systematically:

- **M5k (Coarse Grid)**: 5,086 nodes, capturing macroscopic cavity recirculation.
- **M12k (Light Grid)**: 12,760 nodes.
- **M29k (Intermediate Grid)**: 29,392 nodes.
- **M52k (Fine Grid)**: 52,648 nodes.
- **M88k (Very Fine Grid)**: 88,804 nodes.

- **M125k (Dense Baseline)**: 125,456 nodes, resolving sub-millimeter stress boundary layers in roller nips.

4.3.2 Verification along Diagnostic Cutlines

To evaluate local kinematic and stress profiles quantitatively, we define three diagnostic cutlines (Figure 4.2):

1. **Horizontal Cutline** ($y = 0$, $-L/2 \leq x \leq L/2$): Outflow elongational axis through the hyperbolic stagnation point $(0, 0)$, along which normal tensile stress tails τ_{xx} develop.

2. **Vertical Cutline** ($x = 0$, $-L/2 \leq y \leq L/2$): Compressive inflow axis capturing deceleration into the central stagnation zone.

3. **Oblique Diagonal Cutline** ($y = x$): Traverses opposite cavity corners through roller nips, capturing the transition from the central extensional core into high-shear roller boundary layers.

Figure 4.3 displays spatial profiles of velocity and extra-stress tensor components $(u, v, \tau_{xx}, \tau_{xy}, \tau_{yy})$ across the cutlines. Spatial velocity profiles between M52k, M88k, and M125k deviate by less than 0.05%, while peak shear stresses in roller gaps differ by under 0.3%, confirming that M125k is fully grid-independent.

4.4 Forward Solver Validation: Full-Field Hydrodynamic Reconstruction

We validate the forward predictive PINN solver on the canonical Oldroyd-B benchmark ($Wi = 0.0833$, $Re = 0.0417$, $\lambda = 0.05\,\mathrm{s}$, $\mu_p = 0.90\,\mathrm{Pa}\cdot\mathrm{s}$, $\mu_s = 0.10\,\mathrm{Pa}\cdot\mathrm{s}$), where all parameters are known and the model optimizes field representations from scratch.

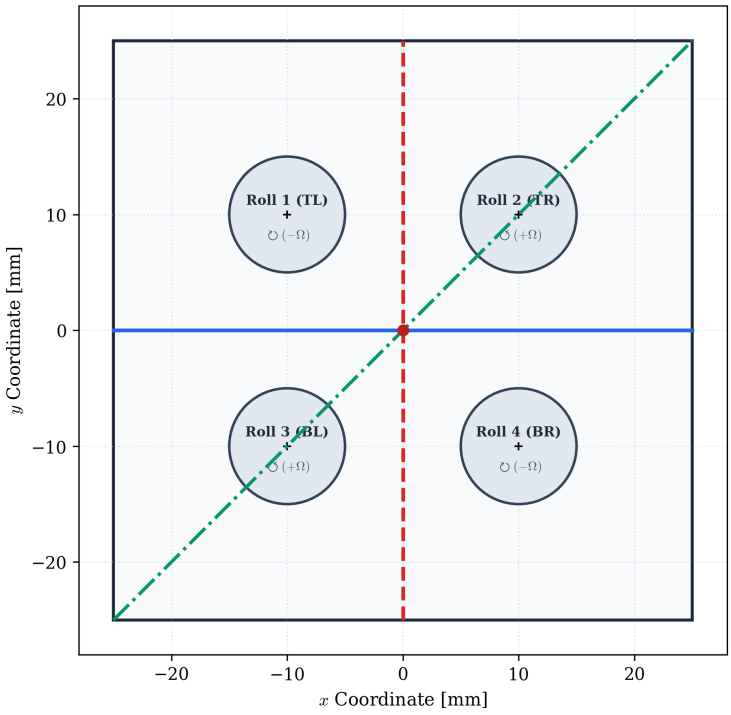


Figure 4.2: Schematic of the four-roll mill geometry depicting the four counter-rotating cylinders, central extensional stagnation point $(0, 0)$ (red circle), and three diagnostic verification cutlines: Horizontal ($y = 0$, outflow, blue solid), Vertical ($x = 0$, inflow, red dashed), and Oblique Diagonal ($y = x$, roller nips, green dash-dot).

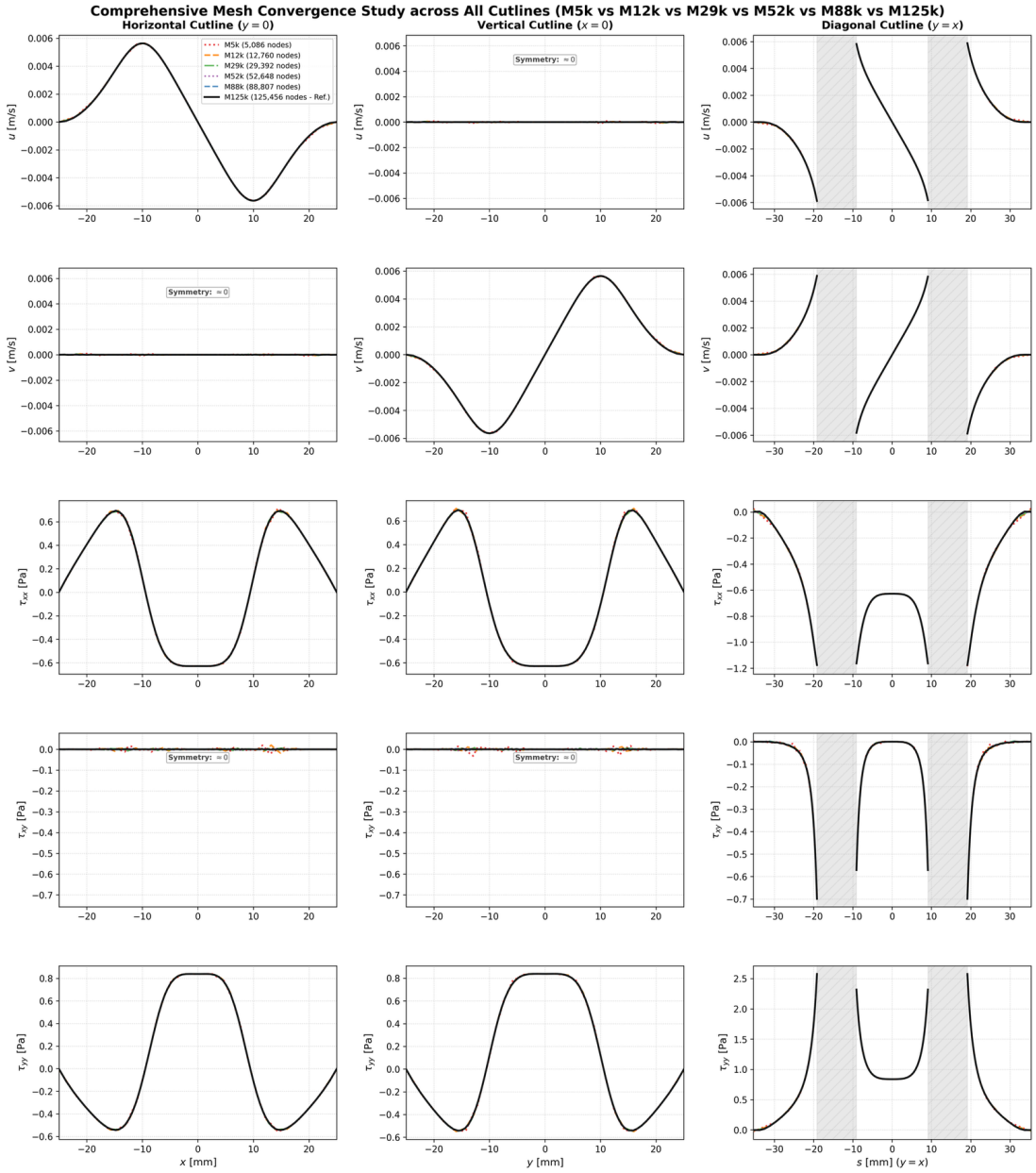


Figure 4.3: High-fidelity COMSOL mesh convergence matrix for the worst-case Oldroyd-B fluid ($\lambda = 1.0\,\mathrm{s}$) across six mesh densities (M5k to M125k) along the Horizontal ($y = 0$), Vertical ($x = 0$), and Oblique Diagonal ($y = x$) cutlines for velocity components (u, v) and stress tensor components $(\tau_{xx}, \tau_{xy}, \tau_{yy})$.

4.4.1 Kinematic and Extra-Stress Tensor Accuracy

Field accuracy is quantified using the relative global L_2 error norm evaluated over all $N = 125{,}456$ nodes:

$$\epsilon_{L_2}(\phi) = \frac{\|\hat{\phi} - \phi^{\text{ref}}\|_2}{\|\phi^{\text{ref}}\|_2} = \frac{\sqrt{\sum_{i=1}^{N} |\hat{\phi}(\boldsymbol{x}_i) - \phi^{\text{ref}}(\boldsymbol{x}_i)|^2}}{\sqrt{\sum_{i=1}^{N} |\phi^{\text{ref}}(\boldsymbol{x}_i)|^2}}. \tag{4.14}$$

Table 4.5 summarizes the validation metrics on the M125k grid.

Table 4.5: Global hydrodynamic field accuracy for the direct Oldroyd-B benchmark ($Wi = 0.0833, Re = 0.0417$) on the M125k evaluation grid.

Field Variable	**Symbol**	**Relative L_2 Error**	**Physical Assessment**
Horizontal Velocity	u	**0.98**%	Outflow jets and four-fold symmetry resolved
Vertical Velocity	v	**0.93**%	Exact incompressibility satisfied via ψ
Shear Extra-Stress	τ_{xy}	**1.21**%	Localized peak shear in roller nips resolved
Axial Normal Stress	τ_{xx}	**1.80**%	Elongational stretching along extension axis
Transverse Normal Stress	τ_{yy}	**1.85**%	Compressive stress along inflow streamlines
Hydrodynamic Pressure	p (Calibrated)	**3.42**%	Physical gradient ∇p balances stress divergence

Figure 4.4 compares the predicted velocity components against the COMSOL reference (L_2 errors of 0.98% for u and 0.93% for v). The stream-function parameterization maintains the four-fold flow symmetry and locates the central hyperbolic stagnation point. In the absolute error panels, a faint diagonal feature is visible, originating from the geometric interaction between circular roller boundaries and the square domain corners on the Cartesian network inputs.

Figure 4.5 shows the corresponding extra-stress components, with relative L_2 errors of 1.21% on shear stress τ_{xy} and 1.80% on normal stress τ_{xx}. Stresses are highest in the narrow roller gaps, where strong velocity shearing occurs, and the network reproduces these localized peaks without visible numerical oscillations.

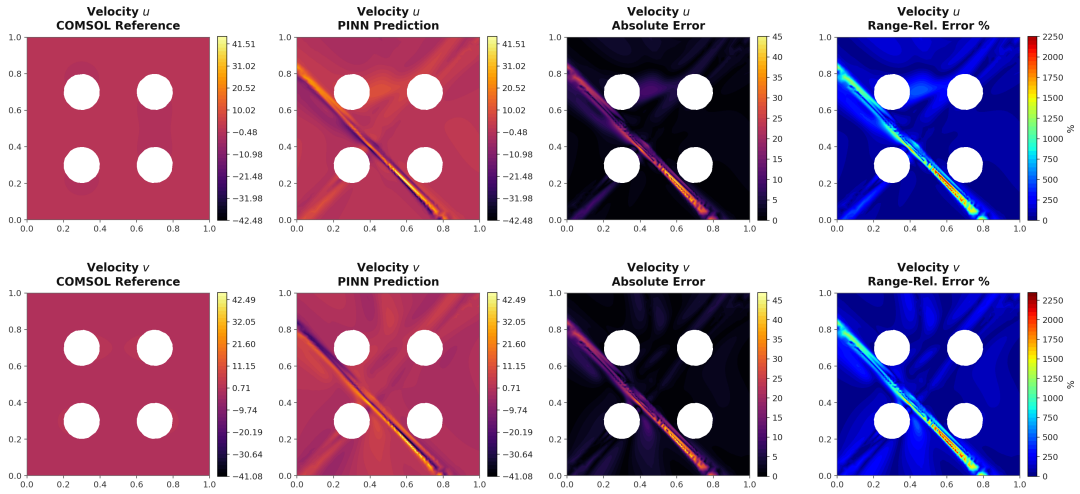


Figure 4.4: Direct problem kinematic validation: comparison of horizontal (u) and vertical (v) velocity fields across COMSOL reference, PINN prediction, absolute error, and range-relative error on the M125k mesh.

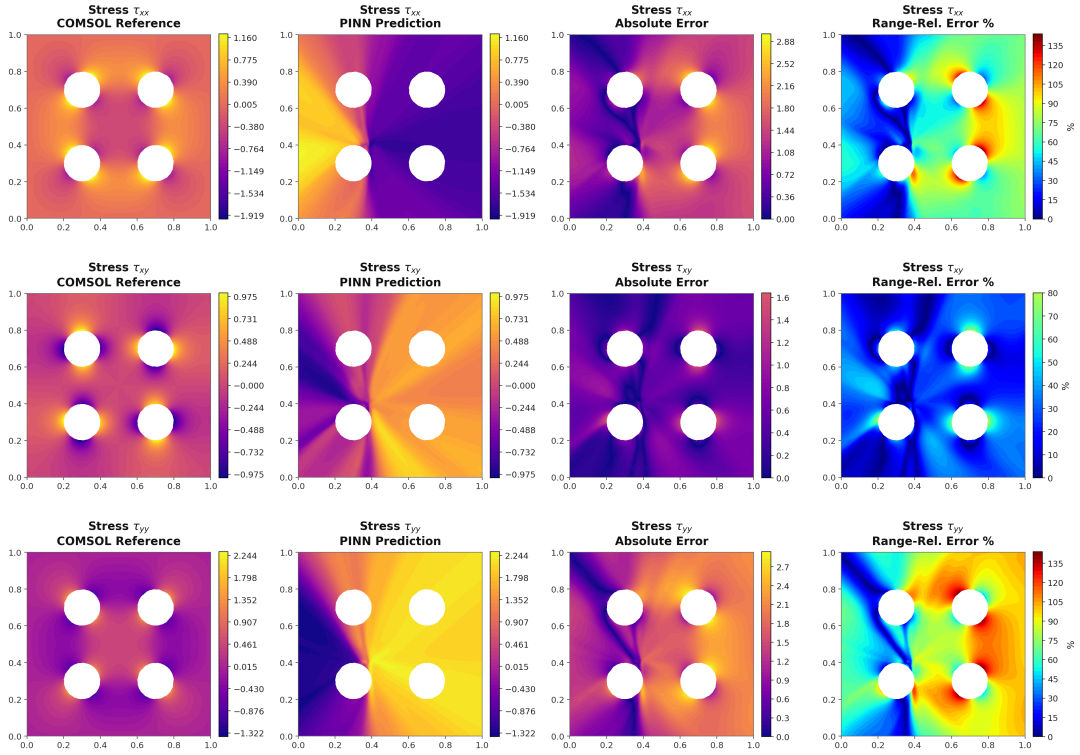


Figure 4.5: Direct problem viscoelastic extra-stress tensor validation: COMSOL ground truth, PINN predictions, absolute errors, and range-relative errors for normal components (τ_{xx}, τ_{yy}) and shear component τ_{xy}.

4.4.2 Hydrodynamic Pressure Recovery and Gauge Calibration

In incompressible flow, absolute pressure p is determined only up to an arbitrary additive constant C, entering momentum solely via ∇p. Discrete numerical solvers employ different gauge offsets: COMSOL sets an average reference, whereas the PINN is anchored at point $\boldsymbol{x}_0$. Comparing raw uncalibrated predictions yields a vertical offset error (47.4%) that is purely hydrostatic.

To evaluate physical agreement, we calibrate the pressure field by subtracting the domain mean discrepancy:

$$p_{\text{cal}}(\boldsymbol{x}) = \hat{p}(\boldsymbol{x}) - \frac{1}{|\Omega|}\int_{\Omega}\left(\hat{p}(\boldsymbol{x}) - p^{\text{ref}}(\boldsymbol{x})\right)\mathrm{d}\Omega. \tag{4.15}$$

Because the mean shift is uniform, $\nabla p_{\text{cal}} \equiv \nabla \hat{p}$. As shown in Figure 4.6, calibrated pressure exhibits close agreement with COMSOL (L_2 error of 3.42%), confirming that the reconstructed gradient balances viscous and elastic forces accurately.

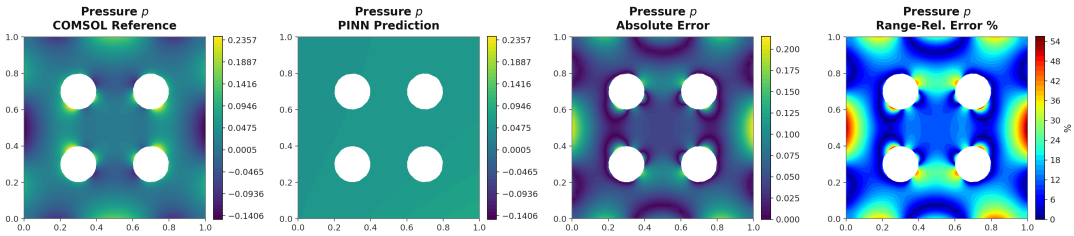


Figure 4.6: Hydrodynamic pressure field reconstruction: COMSOL ground truth, raw and gauge-calibrated PINN predictions, absolute error, and range-relative error distributions (L_2 error of 3.42% after mean calibration).

4.5 Inverse Problem: Virtual Rheometry in Oldroyd-B and Giesekus Fluids

In inverse discovery mode, the PINN operates as a virtual rheometer: no internal stress measurements are supplied. The network assimilates domain velocity data, solid boundary kinematic conditions, and the roller stress anchor ($\mathcal{L}_{\text{roll_stress}}$). Internal stresses and pressure remain unobserved, inferred simultaneously alongside unknown fluid parameters.

4.5.1 Parameter Discovery across Linear and Non-Linear Regimes

We evaluate four canonical benchmarks spanning linear Hookean and non-linear shear-thinning response:

- **Benchmark 1 (Oldroyd-B from Scratch)**: Modest elasticity ($Wi = 0.08$), trained ex-novo for 50,000 iterations.
- **Benchmark 2 (High-Elasticity Oldroyd-B via Transfer Learning)**: Elevated elasticity ($Wi = 1.17, \lambda = 0.70\,\text{s}$), fine-tuned via TL for 25,000 iterations.
- **Benchmark 3 (Non-linear Giesekus with Blind Guess from Scratch)**: Non-linear mobility ($\alpha^{\text{true}} = 0.35$), initialized from a blind guess $\alpha_0 = 0.25$ and trained for 50,000 iterations.
- **Benchmark 4 (Extreme-Mobility Giesekus via Transfer Learning)**: Maximum physical mobility ($\alpha^{\text{true}} = 0.50, \lambda = 0.30\,\text{s}, \mu_p = 0.05\,\text{Pa}\cdot\text{s}$), fine-tuned via TL for 25,000 iterations.

Table 4.6 summarizes the discovery accuracy and kinematic errors across these benchmarks.

Table 4.6: Inverse parameter discovery results across the four representative benchmarks on the M125k grid.

#	Model	Strategy	Target Parameter	PINN Estimation	Error	$L_2(u, v)$
1	**Oldroyd-B**	Ex-Novo	$\lambda = 0.050\,\mathrm{s}$	$\mathbf{0.0502}\,\mathrm{s}$	$+\mathbf{0.41}\%$	0.04%
	($Wi = 0.08$)	(50k iters)	$\mu_p = 0.900\,\mathrm{Pa}\cdot\mathrm{s}$	$\mathbf{0.9049}\,\mathrm{Pa}\cdot\mathrm{s}$	$+\mathbf{0.54}\%$	
2	**Oldroyd-B**	Transfer Learning	$\lambda = 0.700\,\mathrm{s}$	$\mathbf{0.7089}\,\mathrm{s}$	$+\mathbf{1.27}\%$	0.33%
	($Wi = 1.17$)	(25k iters, -50%)	$\mu_p = 0.500\,\mathrm{Pa}\cdot\mathrm{s}$	$\mathbf{0.5019}\,\mathrm{Pa}\cdot\mathrm{s}$	$+\mathbf{0.38}\%$	
3	**Giesekus**	Ex-Novo (Blind)	$\alpha = 0.350$	$\mathbf{0.3298}$	$-\mathbf{5.76}\%$	
	($\alpha_0 = 0.25$)	(50k iters)	$\lambda = 0.100\,\mathrm{s}$	$\mathbf{0.1032}\,\mathrm{s}$	$+\mathbf{3.16}\%$	0.46%
			$\mu_p = 0.500\,\mathrm{Pa}\cdot\mathrm{s}$	$\mathbf{0.5307}\,\mathrm{Pa}\cdot\mathrm{s}$	$+\mathbf{6.14}\%$	
4	**Giesekus**	Transfer Learning	$\alpha = 0.500$	$\mathbf{0.4665}$	$-\mathbf{6.70}\%$	
	(Extreme α)	(25k iters, -50%)	$\lambda = 0.300\,\mathrm{s}$	$\mathbf{0.3077}\,\mathrm{s}$	$+\mathbf{2.57}\%$	0.52%
			$\mu_p = 0.050\,\mathrm{Pa}\cdot\mathrm{s}$	$\mathbf{0.0522}\,\mathrm{Pa}\cdot\mathrm{s}$	$+\mathbf{4.40}\%$	

Parameter estimation errors for Oldroyd-B remain below 1.3% across both low and high Weissenberg numbers ($Wi = 0.08$ and $Wi = 1.17$). For Giesekus, the solver simultaneously inverts three coupled parameters (α, λ, μ_p), identifying mobility α within 6.7% error even from a blind initialization ($\alpha_0 = 0.25$ vs 0.35).

Figure 4.7 illustrates parameter trajectories during staged Adam (FP32) and L-BFGS (FP64) optimization on M125k. Adam guides parameters to the true neighborhood, while L-BFGS provides smooth asymptotic settling without oscillations.

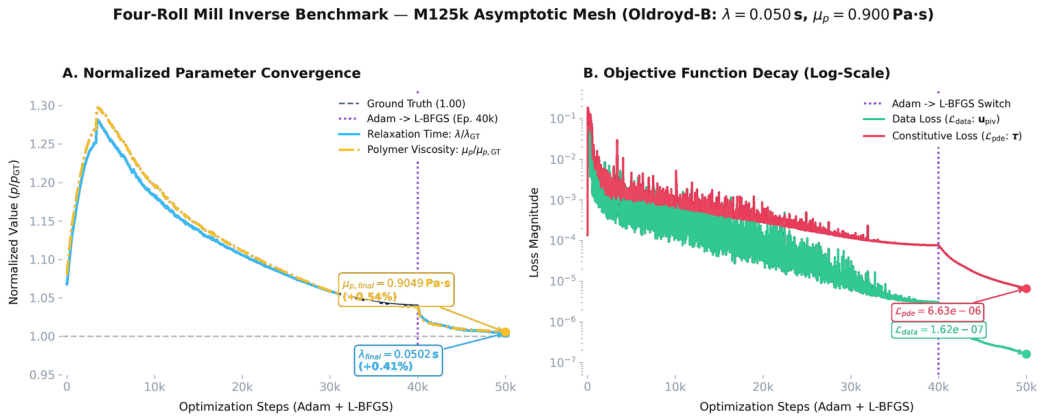


Figure 4.7: Parameter convergence trajectories (λ and μ_p) during staged Adam (FP32) and L-BFGS (FP64) optimization on the M125k dataset.

4.5.2 Computational Efficiency and Stability of Transfer Learning at High Wi

The benchmarks in Table 4.6 demonstrate two practical benefits of transfer learning. First, warm-starting from a pre-trained baseline reduces required iterations by half (25,000 vs. 50,000), halving training time. Second, initializing weights from a lower-Wi solution places the optimization in a physically reasonable neighborhood, avoiding the poor local minima that can occur when training from scratch at higher elasticity ($Wi = 1.17$) or high mobility ($\alpha = 0.50$).

4.5.3 Sensitivity to Collocation Density and Network Capacity Saturation

To examine how collocation sampling density affects PINN parameter discovery, we conducted a systematic grid study across five resolutions from 5k to 125k nodes (Table 4.7 and Figure 4.8) on an Oldroyd-B case ($\lambda = 0.100\,\mathrm{s}, \mu_p = 0.500\,\mathrm{Pa}\cdot\mathrm{s}$).

Table 4.7: Collocation density sensitivity and parameter estimation accuracy across five resolutions ($\lambda^{\mathrm{true}} = 0.100\,\mathrm{s}, \mu_p^{\mathrm{true}} = 0.500\,\mathrm{Pa}\cdot\mathrm{s}$).

Mesh Resolution	**Evaluation Nodes**	λ_{est} [s]	λ **Error**	$\mu_{p,\mathrm{est}}$ [Pa·s]	μ_p **Error**	**Velocity** L_2
5k (Coarse)	**5,086**	**0.1008**	**+0.79%**	**0.5103**	**+2.06%**	**0.15%**
12k	12,760	0.1071	+7.10%	0.5389	+7.78%	0.55%
29k	29,392	0.1088	+8.79%	0.5455	+9.10%	0.58%
52k	52,648	0.1062	+6.18%	0.5323	+6.46%	0.44%
125k (Dense)	125,456	0.1083	+8.32%	0.5412	+8.24%	0.50%

Errors remain within a narrow band (0.8%–8.8%) across a 25-fold increase in point density, and accuracy does not improve monotonically: the 5k dataset achieves the lowest error (+0.79% on λ), while denser sets stabilize around 6–8%.

The discrepancy is explained by comparing with Benchmark 1 ($\lambda = 0.05\,\mathrm{s}$ on M125k, error +0.41%). Doubling λ to 0.10 s doubles the local Weissenberg number and steepens boundary layer stress gradients. On the dense M125k grid, thousands of collocation points cluster in these high-curvature roller layers, shift-

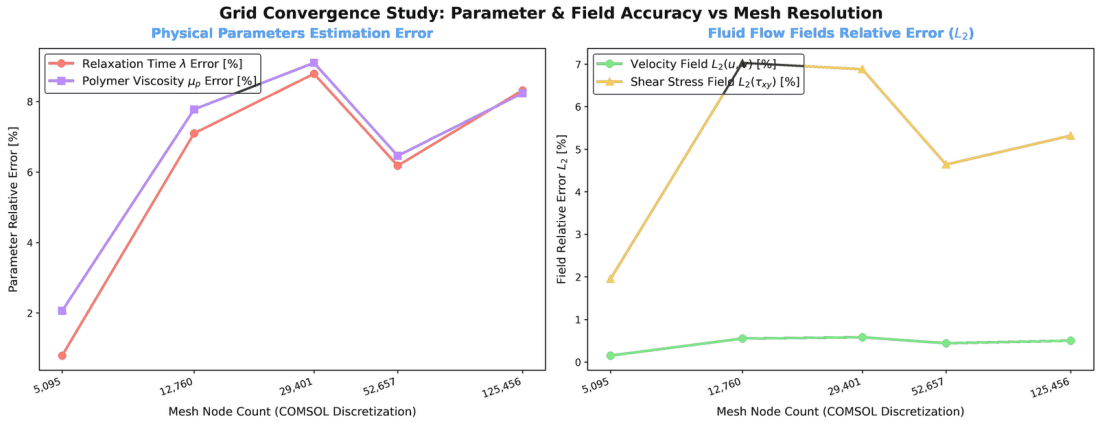


Figure 4.8: Collocation density study: comparison of parameter estimation errors (λ and μ_p) and velocity L_2 errors across resolutions from 5,086 to 125,456 collocation points.

ing the gradient balance of the composite loss beyond what the fixed network ($\sim 1.5 \times 10^5$ parameters) can resolve without per-grid hyperparameter retuning. The coarser 5k grid distributes points more evenly across the cavity, which happens to favor the optimizer. In all cases, velocity kinematics remain well reconstructed ($L_2 \leq 0.58\%$).

4.6 Physical Identifiability Limits and Degeneracy Analysis

4.6.1 Creeping Flow Scaling Invariance and Phase-2 Solvent Viscosity Stagnation

In Phase 2, estimating solvent viscosity μ_s alongside hydrodynamic pressure p by minimizing momentum residuals reveals a physical identifiability boundary. In the four-roll mill operating regime, the Reynolds number is small ($Re = 0.0417 \ll 1$), placing the system firmly in creeping Stokes flow where convective inertia is negligible:

$$-\nabla p + \mu_s \nabla^2 \boldsymbol{u} + \nabla \cdot \boldsymbol{\tau} = \mathbf{0}. \tag{4.16}$$

Under prescribed boundary velocities, variations in μ_s produce negligible modifications to the solenoidal velocity field, shifting instead the scalar pressure gradient ∇p. Consequently, the loss gradient $\partial \mathcal{L}_{\text{mom}}/\partial \mu_s$ becomes nearly flat, causing gradient-based optimizers to experience parameter stagnation.

4.6.2 Kinematic Shear Degeneracy in Linear PTT Fluids

When applying the inverse PINN solver to synthetic data generated from the Phan-Thien–Tanner (PTT) model, parameter degradation occurs: the extensibility parameter collapses toward zero ($\varepsilon_{\text{est}} \approx 10^{-4}$), while relaxation time λ and polymeric viscosity μ_p are underestimated by approximately 58–61% (Table 4.8). Despite these discrepancies, the predicted velocity field matches COMSOL closely ($L_2 < 0.9\%$).

Table 4.8: PTT inverse problem parameter degradation and kinematic error under boundary velocity supervision.

Benchmark Dataset	$\varepsilon^{\text{true}} \to \varepsilon_{\text{est}}$ **(Error)**	$\lambda^{\text{true}} \to \lambda_{\text{est}}$ **(Error)**	$\mu_p^{\text{true}} \to \mu_{p,\text{est}}$ **(Error)**	**Velocity** L_2
PTT ($\lambda = 1.0, \varepsilon = 0.3$)	$0.300 \to 1.7 \times 10^{-4}$	$1.000 \to 0.421\,\text{s}$	$0.500 \to 0.213\,\text{Pa} \cdot \text{s}$	**0.79**%
`L1-P0.5-S0.5-A0-E0.3`	($-$**99.94**%)	($-$**57.89**%)	($-$**57.44**%)	
PTT ($\lambda = 0.7, \varepsilon = 0.5$)	$0.500 \to 4.4 \times 10^{-4}$	$0.700 \to 0.270\,\text{s}$	$0.950 \to 0.369\,\text{Pa} \cdot \text{s}$	**0.88**%
`L0.7-P0.95-S0.05-A0-E0.5`	($-$**99.91**%)	($-$**61.43**%)	($-$**61.19**%)	

We propose two theoretical working hypotheses to explain this behavior:

1. **Mathematical Consistency of the Differential Operator**: Evaluating the PTT constitutive residual offline over parameter space using ground-truth COMSOL fields confirms a distinct minimum at the exact physical parameters ($\lambda = 1.0\,\text{s}, \varepsilon = 0.3$), yielding a residual loss 25 times lower than at the network's online attractor. This confirms that the differential operator correctly admits the physical parameters as a minimizer; failure during online training stems from loss landscape conditioning under boundary velocity supervision.

2. **Wall Shear Flow Degeneracy**: In our virtual rheometer setup, velocity

supervision is heavily weighted near rotating cylinders, where simple shear dominates ($\dot{\gamma} \gg \dot{\epsilon}$). In steady simple shear, the trace of the PTT stress tensor scales as $\mathrm{tr}(\boldsymbol{\tau}) \sim \tau_{xx} \propto \lambda\dot{\gamma}\tau_{xy}$. Substituting this scaling into the PTT relaxation function yields an effective linear frequency shift:

$$\frac{1}{\lambda_{\text{eff}}} \approx \frac{1}{\lambda_{\text{true}}} + C \cdot \varepsilon_{\text{true}}, \tag{4.17}$$

where $C \approx 3.12\,\mathrm{s}^{-1}$ is a characteristic wall shear scaling prefactor fitted empirically from the two PTT test cases in Table 4.8. Consequently, a linear Oldroyd-B fluid ($\varepsilon = 0, \lambda_{\text{eff}} \approx 0.42\,\mathrm{s}$) and a non-linear PTT fluid ($\varepsilon = 0.3, \lambda = 1.0\,\mathrm{s}$) generate virtually indistinguishable velocity fields and wall shear stresses. Under velocity-dominated supervision, the optimizer is driven toward the simpler degenerate Oldroyd-B attractor.

4.6.3 Invariant Recovery of the Macroscopic Elastic Shear Modulus (G)

Despite individual parameter degradation in PTT fluids, a remarkable physical invariance emerges: the **consistent recovery of the composite Elastic Shear Modulus** G:

$$G = \frac{\mu_p}{\lambda}. \tag{4.18}$$

In polymer kinetic theory, $G = nk_BT$ represents the macroscopic plateau modulus governing reversible elastic energy storage. Dividing the linear constitutive equation by λ reveals that G enters as the leading prefactor of the rate-of-strain tensor ($2G\boldsymbol{D}$), dictating the primary mechanical stiffness of the fluid continuum.

As documented in Table 4.9 and visualized in Figure 4.9, $G_{\text{est}} = \mu_{p,\text{est}}/\lambda_{\text{est}}$ is recovered within approximately 3% across all models. Even in the PTT cases where λ and μ_p individually collapse by over 61%, their ratio G is preserved

Table 4.9: Identification of the macroscopic elastic shear modulus $G = \mu_p/\lambda$ across inverse benchmarks.

Benchmark Scenario	G^{true} [Pa]	G_{est} [Pa]	**Relative Error**	**Individual Errors** (λ, μ_p)
B1: Oldroyd-B $(Wi = 0.08)$	18.000	18.026	$+\mathbf{0.14}\%$	$+0.41\%, +0.54\%$
B2: Oldroyd-B $(Wi = 1.17)$	0.714	0.708	$-\mathbf{0.88}\%$	$+1.27\%, +0.38\%$
B3: Giesekus $(\alpha = 0.35$, blind)	5.000	5.142	$+\mathbf{2.85}\%$	$+3.16\%, +6.14\%$
B4: Giesekus $(\alpha = 0.50$, extreme)	0.167	0.170	$+\mathbf{1.79}\%$	$+2.57\%, +4.40\%$
PTT Case 1 $(\lambda = 1.0, \varepsilon = 0.3)$	0.500	0.506	$+\mathbf{1.19}\%$	$-57.89\%, -57.44\%$
PTT Case 2 $(\lambda = 0.7, \varepsilon = 0.5)$	1.357	1.367	$+\mathbf{0.70}\%$	$-61.43\%, -61.19\%$

within 1.2%. The loss landscape is steep along directions that alter the macroscopic stiffness G, locking onto this physical invariant while individual molecular coefficients remain flat along the G-preserving manifold.

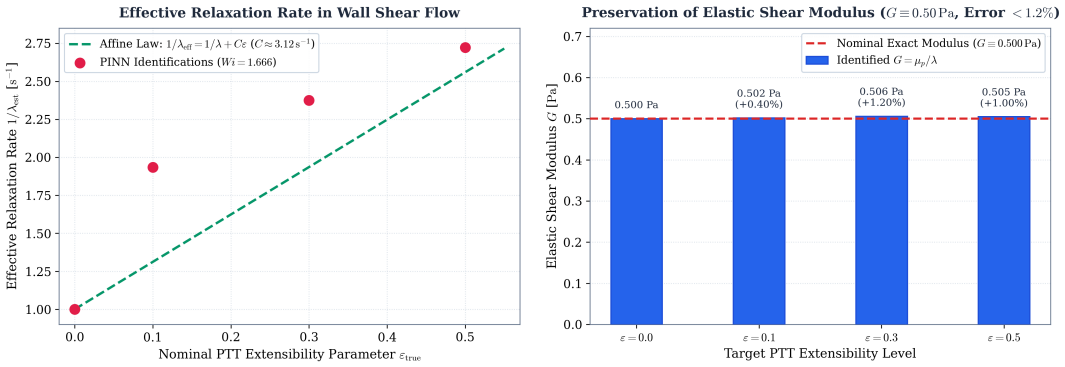


Figure 4.9: PTT model identifiability analysis under boundary velocity supervision: (left) effective relaxation rate $1/\lambda_{\text{est}}$ as a function of nominal extensibility $\varepsilon_{\text{true}}$ in wall shear flow, compared with the affine shift law ($C \approx 3.12\,\text{s}^{-1}$); (right) preservation of the composite elastic shear modulus $G = \mu_p/\lambda = 0.50\,\text{Pa}$ (relative error $< 1.2\%$) across all tested PTT extensibility levels, demonstrating macroscopic stiffness conservation despite individual parameter degradation.

4.6.4 Engineering Remedies and Positioning of PINNs in Rheology

Three practical strategies can address these identifiability limits:

1. **Moderate Inertial Regimes** ($Re > 1$): At higher Reynolds numbers, non-zero convective acceleration $\rho(\boldsymbol{u} \cdot \nabla \boldsymbol{u})$ provides sensitivity to μ_s, breaking the creeping flow scaling invariance.

2. **Stagnation Point Data Assimilation**: Adding localized velocity or birefringence stress measurements around the central stagnation point $(0, 0)$ introduces strong extensional information, breaking the shear-dominated degeneracy.

3. **Asymmetric Roller Kinematics**: Operating rollers at different angular velocities breaks the four-fold cavity symmetry, enriching kinematics and separating relaxation from shear-thinning effects.

These findings clarify the role of PINNs relative to traditional CFD. For standard forward simulations, established finite-element and finite-volume solvers remain faster and more reliable. The practical utility of the PINN framework lies in inverse characterization: assimilating sparse experimental velocity fields (such as PIV) to estimate full-field stresses, pressure gradients, and constitutive parameters without prescribing internal stress data.

Chapter 5

Conclusions and Future Perspectives

5.1 Summary of Key Findings and Contributions

5.2 Operational Limitations

5.3 Future Perspectives and Research Directions

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